

Derived categories and Brauer groups of log del Pezzo surfaces with $\frac{1}{3}(1, 1)$ singularities

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Abstract

Let X be a log del Pezzo surface with only $\frac{1}{3}(1, 1)$ -points, let $\mathcal{X}$ be its canonical stack and $\tilde{X}$ its minimal resolution. Corti–Heuberger organize these surfaces into six cascades of smooth-point blow-ups of rigid bases $Z_1, \dots, Z_6$. We construct marked resolution models and full exceptional collections on the canonical stacks of all six bases and transport them along the cascades, obtaining an explicit full exceptional collection of length $12 - K_X^2 + 4k/3$ on $\mathcal{X}$ for every index-one surface with k singular points. On the coarse surface we compute the Brauer group throughout the classification: it vanishes for $k \leq 5$ and equals $\mathbb{Z}/3$ for $k = 6$. Consequently Karmazyn–Kuznetsov–Shinder descent is untwisted in the one- and three-point cascades and necessarily Brauer-twisted in the six-point cascade, and we give the resulting semiorthogonal decompositions. For quasismooth degree-10 hypersurfaces in $\mathbb{P}(1, 2, 3, 5)$ both sides are written out completely: a length-13 stack collection with its Euler form, a coarse decomposition with one Kalck–Karmazyn component, and an identification of general hypersurfaces with blow-ups of $\mathbb{P}(1, 1, 3)$ at eight general points. Finally, for rational surfaces with an arbitrary basket of $\frac{1}{n_i}(1, 1)$ -points, we express the Euler form of the stack collection in block form in terms of the resolution collection and its intersection numbers with the exceptional curves.

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1 Introduction

Let X be a log del Pezzo surface over $\mathbb{C}$, let $\pi: \mathcal{X} \rightarrow X$ be its canonical stack, and let $\nu: \tilde{X} \rightarrow X$ be its minimal resolution. The three categories

$$D^b(\mathrm{coh} \tilde{X}), \quad D^b(\mathrm{coh} \mathcal{X}), \quad D^b(\mathrm{coh} X)$$

record the quotient singularities of X in three different ways. On the resolution they are invisible. On the stack, each singular point contributes local exceptional objects supported on its residual gerbe, by the special McKay correspondence of Ishii–Ueda [17]: there is a fully faithful functor $\Phi: D^b(\mathrm{coh} \tilde{X}) \rightarrow D^b(\mathrm{coh} \mathcal{X})$ whose semiorthogonal

complement is generated by $n_i - 2$ objects at each point of type $\frac{1}{n_i}(1, 1)$. On the coarse surface, when the descent of Karmazyn–Kuznetsov–Shinder (KKS) [19] applies, the singular points are encoded by derived categories of finite-dimensional local algebras. For a rational surface with quotient singularities, in particular for a log del Pezzo surface (Proposition 2.2), Orlov’s blow-up formula and Ishii–Ueda give a full exceptional collection on its canonical stack; this is observed by Gugiatti–Rota [13, Remark 4.8] and recalled in Proposition 3.5. The purpose of this paper is to make both the stack collections and the coarse decompositions explicit for the surfaces with $\frac{1}{3}(1, 1)$ -points classified by Corti–Heuberger [7], and to identify the obstruction that separates the two sides.

Corti–Heuberger group the non-smooth del Pezzo surfaces with only $\frac{1}{3}(1, 1)$ -points into 29 qG-deformation families and six cascades: every index-one family $X_{k,d}$, with k singular points and degree $d = K_X^2$, is obtained from a rigid base Z_k by blowing up $K_{Z_k}^2 - d$ smooth points [7, Theorems 4 and 6, Corollary 8]. Here $Z_1 = \mathbb{P}(1, 1, 3)$, and for $2 \leq k \leq 6$ the base Z_k is the unique surface in its family; their invariants are collected in Table 1. Because the cascade morphisms are smooth-point blow-ups, all categorical and Brauer-theoretic input is concentrated on these six bases.

To specify the stack collections, we use *marked blow-up presentations* of the base resolutions (Definition 5.3). These record the ordered centers, including infinitely near ones, their incidences with earlier exceptional curves, and the curves contracted to the singular points. Our first result gives such presentations for the six bases, together with the Orlov collections they determine, and propagates them along the cascades.

Theorem 1.1 (Explicit collections in the Corti–Heuberger cascades). *For each rigid base Z_k , $1 \leq k \leq 6$, the minimal resolution $\tilde{Z}_k$ admits a marked presentation by point blow-ups of the surface Y_0 , with the number of blow-ups specified in Table 1. The associated Orlov full exceptional collection is $\mathcal{E}_k = \langle A_1^{(k)}, \dots, A_{N_k}^{(k)} \rangle$, where $N_k = \text{rk } K_0(\tilde{Z}_k)$ is recorded in the same table (Proposition 5.4, Appendix C).*

Let $X \in X_{k,d}$ and choose a marked Corti–Heuberger presentation $X \rightarrow Z_k$ by $s = K_{Z_k}^2 - d$ smooth points of Z_k ; these are distinct by ampleness, as shown in (3). Let $\gamma: \tilde{X} \rightarrow \tilde{Z}_k$ be the induced map of minimal resolutions, with disjoint exceptional curves $D_1, \dots, D_s$. Then

$$\langle \mathcal{O}_{D_s}(-1), \dots, \mathcal{O}_{D_1}(-1), L\gamma^* \mathcal{E}_k \rangle$$

is a full exceptional collection on $D^b(\text{coh } \tilde{X})$, and prepending the k residual-gerbe objects to its image under Φ gives a full exceptional collection on $D^b(\text{coh } \mathcal{X})$ of length

$$N_k + s + k = 12 - d + \frac{4k}{3}.$$

The base constructions occupy Appendix C; the propagation is Corollary C.3. The three- and six-point bases are handled by toric contractions, the four-point base requires an elementary transformation of $\mathbb{F}_1$ (Proposition C.2), and the two- and five-point bases use explicit incidence data and the pentagon construction of [7].

Passing from the resolution to the singular coarse surface requires compatibility with the contraction. Under this condition, KKS descend a semiorthogonal decomposition and identify the descended components when the blocks over the contracted curves are *adherent* [19, Theorems 2.12 and 3.16]. An untwisted adherent decomposition forces the divisor class group $\text{Cl}(X)$ to be torsion-free [19, Corollary 4.8], and Bright’s exact sequence [4] identifies this torsion, dually, with the Brauer group $\text{Br}(X) \cong \text{coker}(\theta: \text{Pic } \tilde{X} \rightarrow \Lambda^\vee)$,

where Λ is generated by the exceptional curves and θ is their intersection map (Corollary 4.9). Since the Brauer group is invariant under smooth-point blow-ups (Lemma B.5), it suffices to compute it on the six bases. This gives our second result.

Theorem 1.2 (Brauer obstruction in the Corti–Heuberger cascades). *Let X belong to a Corti–Heuberger cascade with k singular points, all of type $\frac{1}{3}(1, 1)$. Then*

$$\mathrm{Br}(X) \cong \begin{cases} 0, & 1 \leq k \leq 5, \\ \mathbb{Z}/3, & k = 6. \end{cases}$$

In particular, the six-point cascade is the unique cascade in which the Brauer group obstructs an untwisted adherent decomposition in the sense of KKS.

The group orders are already determined by Corti–Heuberger’s defect σ [7, Section 2.1, Proposition 18(b), Lemma 28]. What we add is the identification of $\mathrm{Br}(X)$ with the dual of their glue group $\Lambda^{\mathrm{sat}}/\Lambda$ (Lemma B.2), explicit intersection maps on toric and k -gon models of the bases, and the alternating generator $\frac{1}{3}(E_1 - E_2 + E_3 - E_4 + E_5 - E_6)$ of the glue group of Z_6 (Appendix B). Vanishing of the Brauer group is necessary for untwisted descent but does not construct the adherent blocks. Those are available when the base is toric, which happens exactly for $k \in \{1, 3, 6\}$ [7, Tables 1, 2 and 4], [22, Table 1], and they propagate through the cascade blow-ups (Lemma 4.4). This yields untwisted KKS decompositions in the one- and three-point cascades (Corollary 5.11 and Corollary B.6, respectively) and a decomposition of a nontrivially Brauer-twisted category in the six-point cascade (Corollary B.8). For $k = 2, 4, 5$ the bases are not toric, their toric degenerations acquire additional singularities, and the existence of simultaneous adherent blocks remains open (Remark B.7).

Both the stack collection and the coarse decomposition can be written out in full for the quasismooth degree-10 hypersurfaces in $\mathbb{P}(1, 2, 3, 5)$, which belong to the family $X_{1,1/3}$. We write $\mathbb{F}_n$ for the n -th Hirzebruch surface, C_0 for its negative section and f for its fiber class.

Theorem 1.3 (The hypersurface X_{10}). *Let $X_{10} \subset \mathbb{P}(1, 2, 3, 5)$ be a quasismooth hypersurface of degree 10, with canonical stack $\mathcal{X}_{10}$ and minimal resolution $\nu: \tilde{X}_{10} \rightarrow X_{10}$. Then X_{10} has a unique singular point p , of type $\frac{1}{3}(1, 1)$, and $\tilde{X}_{10}$ is the blow-up $\sigma: \tilde{X}_{10} \rightarrow \mathbb{F}_3$ at eight distinct points off C_0 , with pairwise disjoint exceptional curves $D_1, \dots, D_8$.*

(i) *$D^b(\mathrm{coh} \mathcal{X}_{10})$ admits the full exceptional collection*

$$\langle \mathcal{E}_p, \Phi(\mathcal{O}_{D_1}(-1)), \dots, \Phi(\mathcal{O}_{D_8}(-1)), \Phi(\sigma^*\mathcal{O}), \Phi(\sigma^*\mathcal{O}(f)), \Phi(\sigma^*\mathcal{O}(C_0 + 3f)), \Phi(\sigma^*\mathcal{O}(C_0 + 4f)) \rangle$$

of length 13, whose Euler form is computed in Proposition 5.9.

(ii) *The coarse surface admits a semiorthogonal decomposition*

$$D^b(\mathrm{coh} X_{10}) = \langle D^b(K(3, 1)\text{-mod}), F_1, \dots, F_{10} \rangle, \\ K(3, 1) \cong \mathbb{C}[z_1, z_2]/(z_1, z_2)^2,$$

with exceptional objects F_i made explicit in Theorem 5.10.

(iii) *A general such hypersurface is the anticanonical model of the blow-up of $\mathbb{P}(1, 1, 3)$ at eight general smooth points, and conversely general centers give a general hypersurface (Propositions A.5 and 5.7).*

The assertion about the unique singular point is proved in [Lemma 5.5](#). The member-wise eight-point presentation follows from Corti–Heuberger [[7](#), Corollary 8(1)] and the distinctness argument ([3](#)). Part (i) is the $k = 1$, $s = 8$ case of [Theorem 1.1](#); part (ii) is the KKS model for $\mathbb{P}(1, 1, 3)$ [[19](#), Example 3.17] propagated through the eight blow-ups ([Corollary 4.6](#)); part (iii) combines the Reid–Suzuki construction [[24](#)] with a dominance argument in [Appendix A](#).

Anticanonically embedded weighted hypersurfaces were classified by Johnson–Kollár [[18](#)]; the collections of Elagin [[11](#)] and Gugiatti–Rota [[13](#)] concern different hypersurface series.

The Euler computation for X_{10} is a specialization of a uniform formula for mixed baskets. For a rational surface with points p_i of type $\frac{1}{n_i}(1, 1)$, $1 \leq i \leq r$, allowing different $n_i \geq 2$, and a full exceptional collection $\langle G_1, \dots, G_N \rangle$ on the resolution, the Euler matrix of the stack collection of [Corollary 3.6](#) is block upper triangular: the diagonal blocks are the local blocks $T_{n_i} = (I - J)^2$, with J the nilpotent shift, and the resolution matrix $\chi(G_a, G_b)$; the off-diagonal block consists of the pairings $\chi(\mathcal{E}_{p_i, j}, \Phi(G_a))$ ([Proposition 6.1](#)). When the G_a are line bundles, these pairings depend only on the intersection numbers $c_1(G_a) \cdot E_i$ with the exceptional curves, so the resolution matrix and the intersection data determine the entire Euler form. The resolution rank and the stack length are

$$N = 12 - K_X^2 + \sum_{i=1}^r \frac{(n_i - 2)^2}{n_i}, \quad \ell = N + \sum_{i=1}^r (n_i - 2),$$

which specialize to $\ell = 12 - K_X^2 + 4r/3$ when all $n_i = 3$ ([Corollary 3.7](#)). The one-point Hirzebruch specialization ([Proposition 6.3](#)) covers the Cavey–Prince cascades from $\mathbb{P}(1, 1, n)$ [[6](#)], including infinitely near centers, and recovers the Euler form of [Theorem 1.3\(i\)](#).

[Sections 2 to 4](#) develop the resolution, stack and descent tools; [Section 5](#) treats the cascades and X_{10} , and [Section 6](#) computes the Euler forms. [Appendices A, B and C](#) contain the generality comparison, the Brauer computations, and the marked base constructions.

Notation

In the general theory, $r = \# \text{Sing}(X)$ and $p_i \sim \frac{1}{n_i}(1, 1)$; for one point we write n . In the Corti–Heuberger notation $X_{k,d}$, Z_k , we have $n_i = 3$ and $r = k$. We distinguish resolution curves E_i over the singular points, blow-up curves D_i over smooth centers, and local objects $\mathcal{E}_{p_i, j}$; for a $\frac{1}{3}(1, 1)$ -point we write $\mathcal{E}_{p_i} = \mathcal{E}_{p_i, 2}$.

2 Resolutions and exceptional collections

We work over $\mathbb{C}$. The starting point for the stack construction in [Section 3](#) is a full exceptional collection on the minimal resolution of a log del Pezzo surface. Rationality allows us to present this resolution as a sequence of point blow-ups of $\mathbb{P}^2$ or a Hirzebruch surface. Orlov’s blow-up formula then transports the standard collection on the starting surface to the resolution, yielding [Corollary 2.7](#).

We begin with the surface geometry, recalling the rationality argument before turning to exceptional collections. The relevant background on log del Pezzo surfaces is developed in [[2](#), section “Basic facts about log del Pezzo surfaces”], and the rationality statement

is also recalled in [13, Remark 4.3]. Our treatment of exceptional collections on smooth rational surfaces follows [15].

Definition 2.1. A *log del Pezzo surface* is a normal projective surface X with klt singularities such that $-K_X$ is ample.

For normal surfaces, klt singularities are precisely quotient singularities [20, Proposition 4.18]. They are also rational [20, Theorem 5.22]. Hence, if $f : \tilde{X} \rightarrow X$ is any resolution, then $f_*\mathcal{O}_{\tilde{X}} = \mathcal{O}_X$ and $R^i f_*\mathcal{O}_{\tilde{X}} = 0$ for all $i > 0$. The Leray spectral sequence therefore gives a canonical isomorphism

$$H^1(\tilde{X}, \mathcal{O}_{\tilde{X}}) \cong H^1(X, \mathcal{O}_X).$$

We recall the rationality argument.

Proposition 2.2. *Let X be a log del Pezzo surface, and let $f : \tilde{X} \rightarrow X$ be its minimal resolution. The surface $\tilde{X}$ is smooth and rational. In particular, X is rational.*

Proof. Since X has klt singularities and $-K_X$ is ample, Kawamata–Viehweg vanishing gives $H^1(X, \mathcal{O}_X) = 0$. Since the singularities of X are rational, the Leray isomorphism above gives $H^1(\tilde{X}, \mathcal{O}_{\tilde{X}}) = 0$.

It remains to show that $H^0(\tilde{X}, \mathcal{O}_{\tilde{X}}(2K_{\tilde{X}})) = 0$. Suppose, to the contrary, that an effective divisor $D \sim 2K_{\tilde{X}}$ exists. The pullback $-f^*K_X$ is nef because $-K_X$ is ample, and hence $(-f^*K_X) \cdot D \geq 0$. On the other hand, write the discrepancy formula as

$$K_{\tilde{X}} = f^*K_X + \sum_i a_i E_i$$

. Since $f^*K_X \cdot E_i = 0$ for every f -exceptional curve E_i , this formula gives

$$(-f^*K_X) \cdot D = (-f^*K_X) \cdot 2K_{\tilde{X}} = -2K_X^2 < 0.$$

This contradiction implies $H^0(\tilde{X}, \mathcal{O}_{\tilde{X}}(2K_{\tilde{X}})) = 0$. Thus, by Castelnuovo’s rationality criterion, $\tilde{X}$ is rational. Since X is birational to $\tilde{X}$, the surface X is rational as well. $\square$

Example 2.3. The weighted projective plane $\mathbb{P}(1, 1, 3)$ has a unique singularity of type $\frac{1}{3}(1, 1)$. Its minimal resolution is $\mathbb{F}_3 = \mathbb{P}_{\mathbb{P}^1}(\mathcal{O} \oplus \mathcal{O}(3))$, whose negative section C_0 , with $C_0^2 = -3$, is the exceptional locus [19, Example 3.17].

Standard collections on the starting smooth surfaces provide the input for Orlov’s point-blow-up formula.

Definition 2.4. Let $\mathcal{T}$ be a triangulated category. An object $G \in \mathcal{T}$ is *exceptional* if $\text{Hom}(G, G) = \mathbb{C}$ and $\text{Ext}^k(G, G) = 0$ for all $k \neq 0$. An ordered sequence $\langle G_1, \dots, G_n \rangle$ is an *exceptional collection* if each G_i is exceptional and $\text{Ext}^\bullet(G_j, G_i) = 0$ for all $j > i$. The collection is *full* if it generates $\mathcal{T}$ as a triangulated category, that is, under shifts and cones.

The projective line has the standard full exceptional collection

$$D^b(\text{coh } \mathbb{P}^1) = \langle \mathcal{O}, \mathcal{O}(1) \rangle,$$

whereas the projective plane has Beilinson’s full exceptional collection [3]

$$D^b(\text{coh } \mathbb{P}^2) = \langle \mathcal{O}, \mathcal{O}(1), \mathcal{O}(2) \rangle.$$

Orlov's projective bundle formula [23, Corollary 2.7], applied to

$$\mathbb{F}_n = \mathbb{P}_{\mathbb{P}^1}(\mathcal{O} \oplus \mathcal{O}(n))$$

together with the collection on $\mathbb{P}^1$, gives a full exceptional collection on $\mathbb{F}_n$. Let C_0 denote the negative section and f the fiber class. With the convention $C_0^2 = -n$, $C_0 \cdot f = 1$, and $f^2 = 0$, the relative tautological line bundle is

$$\mathcal{O}_{\mathbb{F}_n}(1) \simeq \mathcal{O}(C_0 + nf).$$

After twisting the projective-bundle decomposition by $\mathcal{O}_{\mathbb{F}_n}(1)$, its two blocks are ordered as

$$D^b(\text{coh } \mathbb{F}_n) = \left\langle p^* D^b(\text{coh } \mathbb{P}^1), p^* D^b(\text{coh } \mathbb{P}^1) \otimes \mathcal{O}(C_0 + nf) \right\rangle,$$

where $p : \mathbb{F}_n \rightarrow \mathbb{P}^1$ is the ruling. Expanding the two blocks in this order gives the full exceptional collection

$$\langle \mathcal{O}, \mathcal{O}(f), \mathcal{O}(C_0 + nf), \mathcal{O}(C_0 + (n+1)f) \rangle.$$

More generally, $\langle \mathcal{O}_{\mathbb{P}^1}(a-n), \mathcal{O}_{\mathbb{P}^1}(a-n+1) \rangle$ generates $D^b(\text{coh } \mathbb{P}^1)$ for every $a \in \mathbb{Z}$. Since $\mathcal{O}(C_0 + af) = \mathcal{O}_{\mathbb{F}_n}(1) \otimes p^* \mathcal{O}_{\mathbb{P}^1}(a-n)$, expanding the second projective-bundle block in this pair gives

$$\langle \mathcal{O}, \mathcal{O}(f), \mathcal{O}(C_0 + af), \mathcal{O}(C_0 + (a+1)f) \rangle, \quad a \in \mathbb{Z}, \quad (1)$$

a full exceptional collection for every value of a .

Theorem 2.5 (Orlov's blow-up formula [23, Theorem 4.3 and Corollary 4.4]). *Let S be a smooth projective surface, and let $\beta : \text{Bl}_p S \rightarrow S$ be the blow-up of S at a point p , with exceptional curve D . There is a semiorthogonal decomposition*

$$D^b(\text{coh } \text{Bl}_p S) = \left\langle \mathcal{O}_D(-1), \beta^* D^b(\text{coh } S) \right\rangle,$$

where $\mathcal{O}_D(-1)$ is regarded as an object of $D^b(\text{coh } \text{Bl}_p S)$ via the inclusion $D \hookrightarrow \text{Bl}_p S$. In particular, if $D^b(\text{coh } S)$ admits a full exceptional collection, the category $D^b(\text{coh } \text{Bl}_p S)$ also admits one.

Corollary 2.6 ([15, Theorem 5.6]). *Every smooth projective rational surface admits a full exceptional collection.*

Proof. Every smooth projective rational surface is obtained from either $\mathbb{P}^2$ or a Hirzebruch surface $\mathbb{F}_n$ by a sequence of blow-ups at points. We induct on the number of blow-ups: the starting surfaces $\mathbb{P}^2$ and $\mathbb{F}_n$ have full exceptional collections, and Theorem 2.5 supplies the induction step by adjoining one exceptional object at each point blow-up. $\square$

Together, Proposition 2.2 and Corollary 2.6 yield the following consequence.

Corollary 2.7. *Let X be a log del Pezzo surface, and let $f : \tilde{X} \rightarrow X$ be its minimal resolution. The category $D^b(\text{coh } \tilde{X})$ admits a full exceptional collection.*

3 Canonical stacks and exceptional collections

This section introduces canonical stacks and the local blocks in the special McKay correspondence for a finite set of singularities of types $\frac{1}{n_i}(1, 1)$, with $n_i \geq 2$ allowed to vary. The global decomposition gives the attributed existence statement [Proposition 3.5](#); the discrepancy and rank calculations give [Corollary 3.6](#). We record the $\frac{1}{3}(1, 1)$ specialization for use in the cascades and the local adjoint formula needed for the Euler matrices.

Definition 3.1. Let X be a normal surface with quotient singularities. The *canonical stack* of X is the smooth Deligne–Mumford stack $\pi : \mathcal{X} \rightarrow X$ whose coarse moduli space is X , whose local models at quotient singularities are the corresponding quotient stacks, and whose structure morphism is an isomorphism over the smooth locus of X . Existence for quotient singularities follows from Vistoli’s construction [[26](#), Proposition 2.8 and its proof]. The stack is unique up to isomorphism over X ; see also the universal property in [[12](#), Theorem 4.6 and Remark 4.9].

At a singular point of type $\frac{1}{n}(1, 1)$, with $n \geq 2$, the local analytic model is $\mathbb{A}^2/\mu_n$, and the canonical stack is locally $[\mathbb{A}^2/\mu_n]$. We fix the character convention on the coordinate ring: $\rho_i(\zeta) = \zeta^i$, and the coordinate functions $x, y \in \mathbb{C}[x, y]$ both have weight ρ_1 . The equivariant Koszul resolution of $\mathcal{O}_0 \otimes \rho_i$ has free terms $\mathcal{O} \otimes \rho_i$, $(\mathcal{O} \otimes \rho_{i+1})^{\oplus 2}$, and $\mathcal{O} \otimes \rho_{i+2}$ in degrees 0, -1 , -2 , respectively. All character labels below use this ring-action convention. In particular, on $\mathcal{U}_3 = [\mathbb{A}^2/\mu_3]$, the form $dx \wedge dy$ has character ρ_2 , so $\omega_{\mathcal{U}_3} \simeq \mathcal{O}_{\mathbb{A}^2} \otimes \rho_2$, whereas the anticanonical bundle has character ρ_1 [[13](#), Section 3, equation (17)]. The local character of $\mathcal{O}_{\mathcal{X}_{10}}(1)$, the anticanonical bundle in our hypersurface example, is ρ_1 , not ρ_2 .

The continued fraction $\frac{n}{1} = [n]$ gives a single exceptional curve $E \simeq \mathbb{P}^1$ with $E^2 = -n$.

For the local terminology of the special McKay correspondence, let $G \subset \mathrm{GL}_2(\mathbb{C})$ be a finite small subgroup, let $R = \mathbb{C}[x, y]$, and let

$$\tau : Y \longrightarrow \mathrm{Spec} R^G$$

be the minimal resolution of the quotient singularity. Here small means that G contains no pseudo-reflections. For an irreducible representation ρ of G , set

$$M_\rho := (R \otimes \rho^\vee)^G.$$

This is a reflexive R^G -module. The associated *full sheaf* on Y is

$$\mathcal{M}_\rho := \tau^* M_\rho / \text{torsion}.$$

These constructions associate indecomposable full sheaves on Y with irreducible representations of G [[17](#), Section 2]. Wunram singles out the *special* full sheaves, characterized by

$$H^1(Y, \mathcal{M}_\rho^\vee) = 0.$$

Equivalently, the corresponding representation ρ is called a *special representation*. A representation is called *non-special* if it is not special. This terminology follows Wunram [[27](#)] and the exposition in [[17](#), Section 2].

Lemma 3.2. *Let X be a normal surface with singular points $p_1, \dots, p_r$, where p_i has type $\frac{1}{n_i}(1, 1)$, with $n_i \geq 2$. At p_i , the special representations of μ_{n_i} are precisely ρ_0 and ρ_1 . The non-special representations are $\rho_2, \dots, \rho_{n_i-1}$, so there are $n_i - 2$ of them. For $n_i = 2$ this list is empty; for $n_i = 3$ it consists of ρ_2 alone.*

Proof. At each point, the Hirzebruch–Jung continued fraction $n_i/1 = [n_i]$ gives the recursion

$$i_0 = n_i, \quad i_1 = 1, \quad i_2 = i_0 - n_i i_1 = 0.$$

Wunram’s cyclic special McKay correspondence identifies the entries preceding the terminal zero with the special character labels, read modulo n_i [17, Theorems 2.3 and 3.2]. These give $\rho_{n_i} = \rho_0$ and ρ_1 ; all remaining characters are non-special [13, Example 2.6]. $\square$

Definition 3.3. Let X be a normal surface with singular points $p_1, \dots, p_r$ of types $\frac{1}{n_i}(1, 1)$, with $n_i \geq 2$, and let $\mathcal{X}$ be its canonical stack. For the residual gerbe

$$\iota_{p_i} : B\mu_{n_i} \hookrightarrow \mathcal{X},$$

define the *local exceptional objects*

$$\mathcal{E}_{p_i, j} := \iota_{p_i*}(\mathcal{O}_{B\mu_{n_i}} \otimes \rho_j), \quad 2 \leq j \leq n_i - 1,$$

and their ordered block

$$\mathcal{B}_{p_i} := \langle \mathcal{E}_{p_i, 2}, \dots, \mathcal{E}_{p_i, n_i-1} \rangle.$$

The block is empty when $n_i = 2$. Locally, $\mathcal{E}_{p_i, j}$ is represented by the equivariant module

$$\mathbb{C}[x, y]/(x, y) \otimes \rho_j,$$

namely $\mathcal{O}_0 \otimes \rho_j$ in the notation of Ishii–Ueda. For a point p of type $\frac{1}{3}(1, 1)$, we retain the abbreviation $\mathcal{E}_p := \mathcal{E}_{p, 2}$.

Theorem 3.4 (Ishii–Ueda [17, Theorem 1.4]). *Let X be a surface with at worst quotient singularities, let $\mathcal{X}$ be its canonical stack, and let $Y \rightarrow X$ be the minimal resolution. There is a fully faithful functor*

$$\Phi : D^b(\mathrm{coh} Y) \rightarrow D^b(\mathrm{coh} \mathcal{X})$$

and a semiorthogonal decomposition

$$D^b(\mathrm{coh} \mathcal{X}) = \langle \mathcal{E}_1, \dots, \mathcal{E}_q, \Phi(D^b(\mathrm{coh} Y)) \rangle,$$

where $\mathcal{E}_1, \dots, \mathcal{E}_q$ form an exceptional collection.

If the singularities are $p_1, \dots, p_r$ of types $\frac{1}{n_i}(1, 1)$, with $n_i \geq 2$, the local description [17, Theorem 3.1] and [13, Example 2.6] gives

$$D^b(\mathrm{coh} \mathcal{X}) = \langle \mathcal{B}_{p_1}, \dots, \mathcal{B}_{p_r}, \Phi(D^b(\mathrm{coh} Y)) \rangle,$$

with the blocks of Definition 3.3 and $q = \sum_{i=1}^r (n_i - 2)$. Blocks at distinct points are mutually orthogonal and can be ordered arbitrarily; within each block the character indices increase from 2 to $n_i - 1$.

We use the order of [17, Theorem 1.4]: the exceptional objects precede the image of Φ . In particular,

$$\mathrm{RHom}(\Phi A, \mathcal{E}_i) = 0.$$

This convention fixes the lower-left zero blocks in the Euler matrices. In the projective setting, Φ has both adjoints: the local objects lie in the kernel of its right adjoint, whereas the left adjoint Ψ computes $\mathrm{RHom}(\mathcal{E}_i, \Phi A)$. Thus the nonzero values of $\Psi(\mathcal{E}_i)$ below are consistent with this order; they measure pairings in the other direction.

The following existence statement is observed in [13, Remark 4.8]; we record it for completeness.

Proposition 3.5 (Existence on rational quotient surfaces). *Let X be a normal projective rational surface with quotient singularities, and let $\mathcal{X}$ be its canonical smooth Deligne–Mumford stack. Then $D^b(\mathrm{coh} \mathcal{X})$ admits a full exceptional collection. In particular, this holds for every log del Pezzo surface.*

Proof. The minimal resolution of X is smooth and rational, so [Corollary 2.6](#) supplies a full exceptional collection. Insert it into the Ishii–Ueda decomposition of [Theorem 3.4](#). The log del Pezzo case follows from [Proposition 2.2](#). $\square$

For the one-curve cyclic singularities, the local blocks also determine an ordered collection and its length.

Corollary 3.6 (Mixed-basket length and ordered collection). *Let X be a normal projective rational surface with singular points $p_1, \dots, p_r$ of types $\frac{1}{n_i}(1, 1)$, with $n_i \geq 2$. Let $\mathcal{X}$ be its canonical stack and $f : \tilde{X} \rightarrow X$ its minimal resolution. With the local blocks $\mathcal{B}_{p_i}$ of [Definition 3.3](#), for any full exceptional collection $\langle G_1, \dots, G_N \rangle$ on $\tilde{X}$, the Ishii–Ueda embedding gives the ordered full exceptional collection*

$$\langle \mathcal{B}_{p_1}, \dots, \mathcal{B}_{p_r}, \Phi(G_1), \dots, \Phi(G_N) \rangle$$

on $\mathcal{X}$. The resolution rank and the stack length are

$$N = \mathrm{rk} K_0(\tilde{X}) = 12 - K_X^2 + \sum_{i=1}^r \frac{(n_i - 2)^2}{n_i},$$

$$\ell = N + \sum_{i=1}^r (n_i - 2),$$

so

$$\ell = 12 - K_X^2 + \sum_{i=1}^r \frac{2(n_i - 1)(n_i - 2)}{n_i}.$$

Proof. The resolution has a full exceptional collection by [Corollary 2.6](#). Write $E_i \subset \tilde{X}$ for the exceptional curve over p_i . Thus $E_i^2 = -n_i$, and the discrepancy formula is

$$K_{\tilde{X}} = f^* K_X - \sum_{i=1}^r \frac{n_i - 2}{n_i} E_i.$$

To verify the coefficient of E_i , write it as a_i and intersect with E_i . Adjunction gives

$$K_{\tilde{X}} \cdot E_i = -2 - E_i^2 = n_i - 2,$$

whereas $f^* K_X \cdot E_i = 0$. Hence $-n_i a_i = n_i - 2$, giving $a_i = -(n_i - 2)/n_i$. The curves E_i are disjoint, so

$$K_{\tilde{X}}^2 = K_X^2 - \sum_{i=1}^r \frac{(n_i - 2)^2}{n_i}.$$

Noether’s formula [[14](#), Appendix A, Example 4.1.2] gives $12\chi(\mathcal{O}_{\tilde{X}}) = K_{\tilde{X}}^2 + c_2(\tilde{X})$. For the smooth rational surface $\tilde{X}$, we have $\chi(\mathcal{O}_{\tilde{X}}) = 1$ and $c_2(\tilde{X}) = 2 + \rho(\tilde{X})$, so $\rho(\tilde{X}) = 10 - K_{\tilde{X}}^2$. Hence

$$N = \mathrm{rk} K_0(\tilde{X}) = 2 + \rho(\tilde{X}) = 12 - K_{\tilde{X}}^2 = 12 - K_X^2 + \sum_{i=1}^r \frac{(n_i - 2)^2}{n_i}.$$

Inserting the resolution collection into [Theorem 3.4](#) gives the displayed collection, with $n_i - 2$ additional objects at each point. Its length is

$$\ell = N + \sum_{i=1}^r (n_i - 2) = 12 - K_X^2 + \sum_{i=1}^r \frac{2(n_i - 1)(n_i - 2)}{n_i}. \quad \square$$

For the singularities used in the Corti–Heuberger cascades, the local blocks are singletons. Setting $n_i = 3$ in [Corollary 3.6](#) gives the following specialization.

Corollary 3.7. *Let X be a log del Pezzo surface whose singularities are $p_1, \dots, p_r$, all of type $\frac{1}{3}(1, 1)$. Let $\mathcal{X}$ be its canonical smooth Deligne–Mumford stack, and let $\tilde{X} \rightarrow X$ be its minimal resolution. Then $D^b(\mathrm{coh} \mathcal{X})$ admits a full exceptional collection of length*

$$\ell = 12 - K_X^2 + \frac{4r}{3}.$$

The resolution collection has length

$$N = \mathrm{rk} K_0(\tilde{X}) = 12 - K_X^2 + \frac{r}{3},$$

and each local block in [Corollary 3.6](#) is the singleton $\mathcal{E}_{p_i} = \mathcal{E}_{p_i, 2}$.

The local left adjoint determines the pairings with the resolution block. Write $Y = \mathrm{Tot}_{\mathbb{P}^1} \mathcal{O}_{\mathbb{P}^1}(-n)$, with projection $p : Y \rightarrow \mathbb{P}^1$ and zero section E , and let ψ be the left adjoint of the local Ishii–Ueda embedding ϕ . Only the characters $2 \leq j \leq n - 1$ belong to the exceptional complement; we compute all characters to make the convention transparent. In the notation of [13, Theorem 3.1], our $\rho_{\mathrm{nat}} = \rho_1^{\oplus 2}$, $\rho_{\mathrm{det}} = \rho_2$, and the nontrivial special character is ρ_1 . Thus $\rho_{\mathrm{det}}^\vee = \rho_{n-2}$, while $\rho_j \otimes \rho_{\mathrm{det}} = \rho_1$ means $j = n - 1$. The dual in $M_\rho = (\mathbb{C}[x, y] \otimes \rho^\vee)^{\mu_n}$ selects the weight- ρ summand; it does not relabel the equivariant skyscraper $\mathcal{O}_0 \otimes \rho$. If characters are instead indexed by the geometric action dual to our ring action, every index is negated modulo n . With these conventions, [13, Example 3.5, equation (26)] gives, for $n \geq 3$,

$$\psi(\mathcal{O}_0 \otimes \rho_i) \simeq \begin{cases} 0, & 0 \leq i \leq n - 3, \\ \mathcal{O}_E(-n)[1], & i = n - 2, \\ \mathcal{O}_E(-n + 1), & i = n - 1. \end{cases} \quad (2)$$

Here $\mathcal{O}_E(t)$ has degree t on $E \simeq \mathbb{P}^1$.

Here is the Koszul calculation with the character residues and line-bundle degrees kept separate. Put $P_a = p^* \mathcal{O}_{\mathbb{P}^1}(a)$ for every $a \in \mathbb{Z}$, and $\mathcal{L}_a = \mathcal{O}_{\mathbb{A}^2} \otimes \rho_a$. The $\mathcal{L}_a$ are periodic in a modulo n ; the P_a are not. In particular, P_{a+n} must not be identified with P_a . The weight-one quotient of $\mathbb{C}x \oplus \mathbb{C}y$ over $\mathbb{P}^1$ is $\mathcal{O}_{\mathbb{P}^1}(1)$; its powers give the tautological character summands P_a for $0 \leq a < n$, so universal-family duality gives $\psi(\mathcal{L}_a) = P_{-a}$ in precisely this range [13, proof of Theorem 3.1, equations (20)–(22)]. Apply ψ to the equivariant Koszul resolution

$$0 \longrightarrow \mathcal{L}_{j+2} \xrightarrow{(-y, x)} \mathcal{L}_{j+1}^{\oplus 2} \xrightarrow{(x, y)} \mathcal{L}_j \longrightarrow \mathcal{O}_0 \otimes \rho_j \longrightarrow 0.$$

Let s_0, s_1 be the homogeneous coordinates on $\mathbb{P}^1$, pulled back to Y , and let $t \in H^0(Y, P_{-n})$ be the section cutting out E . Write $u = (-s_1, s_0)^\top$ and $v = (s_0, s_1)$. After reducing the

character indices *before* replacing $\mathcal{L}_a$ by P_{-a} , the complexes, in degrees $-2, -1, 0$, are explicitly

$$\begin{aligned} 0 \leq j \leq n-3 : & \quad [P_{-j-2} \xrightarrow{u} P_{-j-1}^{\oplus 2} \xrightarrow{v} P_{-j}], \\ j = n-2 : & \quad [P_0 \xrightarrow{tu} P_{1-n}^{\oplus 2} \xrightarrow{v} P_{2-n}], \\ j = n-1 : & \quad [P_{-1} \xrightarrow{u} P_0^{\oplus 2} \xrightarrow{tv} P_{1-n}]. \end{aligned}$$

The factor t records multiplication across the character $n-1$ -to- 0 boundary; its degree is $-n$, not zero. The first complex is the exact Euler sequence. In the second, $\ker v = P_{-n}$, so the remaining quotient is $P_{-n}/tP_0 = \mathcal{O}_E(-n)$ in degree -1 . In the third, $\operatorname{coker} u = P_1$, and the induced map $P_1 \xrightarrow{t} P_{1-n}$ is injective with cokernel $\mathcal{O}_E(1-n)$ in degree zero. This proves (2) with the stated shifts. For several singular points the calculation applies componentwise. The restriction $n \geq 3$ here concerns the Euler complement; the descent argument in Section 4 also allows $n = 2$.

We will also use the images of the local generators. For the special characters $j = 0, 1$, these free sheaves belong to the image of ϕ [17, Proposition 1.1], so $\phi(P_0) = \mathcal{L}_0$ and $\phi(P_{-1}) = \mathcal{L}_1$. Applying ϕ to the Euler sequence $0 \rightarrow P_{-1} \rightarrow P_0^{\oplus 2} \rightarrow P_1 \rightarrow 0$ yields

$$\phi(P_1) \simeq \operatorname{coker}(\mathcal{L}_1 \xrightarrow{(-y, x)} \mathcal{L}_0^{\oplus 2}) \simeq (x, y) \otimes \rho_{n-1}.$$

The coordinate maps are those of the universal cluster. In particular, this image is a torsion-free sheaf, not a vector bundle at the origin.

4 The singular surface and the KKS approach

We next pass from the canonical stack to the singular surface X . For a single $\frac{1}{n}(1, 1)$ -point, we start with a decomposition on $\mathbb{F}_n$ compatible with contraction of its negative section and apply the Karmazyn–Kuznetsov–Shinder (KKS) descent and identification theorems. All KKS theorem and equation numbers below refer to the published JEMS version [19]. Compatibility and untwisted adherence persist under point blow-ups away from the section, including infinitely near centers. The propagation argument also works simultaneously for several disjoint exceptional curves, provided the adherent blocks already exist. The Corti–Heuberger application in Section 5 uses $n = 3$.

The stack-theoretic collection does not by itself provide a decomposition of $D^b(\operatorname{coh} X)$; the passage to the coarse surface requires a separate descent argument. The KKS descent method applies to surfaces with rational singularities, a hypothesis satisfied here by the klt property, as recalled in Section 2.

We begin with the compatibility condition that permits descent from a resolution.

Definition 4.1 (KKS compatibility, [19, Definition 2.7]). Let $\tau : \tilde{X} \rightarrow X$ be a resolution of a normal projective surface with rational singularities, viewed as a birational contraction, and let

$$D^b(\operatorname{coh} \tilde{X}) = \langle \tilde{\mathcal{A}}_1, \dots, \tilde{\mathcal{A}}_t \rangle$$

be a semiorthogonal decomposition. The decomposition is *compatible with the contraction* τ if, for every irreducible curve E contracted by τ , the object $\mathcal{O}_E(-1)$ belongs to one of the components $\tilde{\mathcal{A}}_i$.

Theorem 4.2 (KKS descent, [19, Theorem 2.12(i)]). *In the setting of Definition 4.1, a compatible decomposition descends uniquely to*

$$D^b(\operatorname{coh} X) = \langle \mathcal{A}_1, \dots, \mathcal{A}_t \rangle, \quad \mathcal{A}_i = R\tau_*(\tilde{\mathcal{A}}_i).$$

Let $\mathcal{D}_i$ be the union of the exceptional curves assigned to $\tilde{\mathcal{A}}_i$. The restriction of $R\tau_*$ induces an equivalence with the Verdier quotient

$$\mathcal{A}_i \simeq \tilde{\mathcal{A}}_i / \langle \mathcal{O}_E(-1) \mid E \subset \mathcal{D}_i \rangle.$$

Here E runs over the irreducible components of $\mathcal{D}_i$. In particular, pushforward is an equivalence on a block with no assigned curves. The descended projection functors have finite cohomological amplitude. Both assertions are in KKS Theorem 2.12(i); part (ii) concerns admissibility under additional crepancy hypotheses.

To identify the descended blocks, we need the stronger condition of adherence. In the one-curve case used here, its untwisted form has the following concrete description.

Definition 4.3 (One-curve case of [19, Definition 3.6]). Let S be a smooth projective surface such that $H^\bullet(S, \mathcal{O}_S) \cong \mathbb{C}$, and let $E \subset S$ be a smooth rational curve with $E^2 = -n$, where $n \geq 2$. A triangulated subcategory $\mathcal{A} \subset D^b(\text{coh } S)$ is *untwisted adherent* to E if there is a line bundle L_0 such that

$$\mathcal{A} = \langle L_0, \mathcal{O}_E(-1) \rangle, \quad L_0 \cdot E = n - 1.$$

By [19, Lemma 3.5, Definition 3.6, and Example 3.7], this is equivalent to a presentation

$$\mathcal{A} = \langle L_0, L_1 \rangle, \quad L_1 = L_0(E), \quad L_0 \cdot E = n - 1.$$

More generally, a twist b replaces $\mathcal{O}_E(-1)$ by $\mathcal{O}_E(-1 + b)$ and the intersection condition by $L_0 \cdot E = n + b - 1$. We use $b = 0$ for ordinary descent and allow nonzero twists for the Brauer-twisted descent in Appendix B.4.

The following lemma propagates compatibility and adherence for all the contracted curves simultaneously.

Lemma 4.4. *Let S be a smooth projective surface and let $C_1, \dots, C_s \subset S$ be pairwise disjoint smooth rational curves with $C_i^2 = -n_i$, where $n_i \geq 2$. Let $\beta : S' \rightarrow S$ be the blow-up of a point outside $\bigcup_i C_i$, and write C'_i for the strict transforms. If a semiorthogonal decomposition of $D^b(\text{coh } S)$ contains components $\mathcal{A}_i$ such that*

$$\mathcal{O}_{C_i}(-1) \in \mathcal{A}_i,$$

then $\mathcal{O}_{C'_i}(-1) \in L\beta^ \mathcal{A}_i$ for every i . More generally, if $H^\bullet(S, \mathcal{O}_S) \cong \mathbb{C}$ and blocks $\mathcal{A}_i$ have b_i -twisted adherent presentations*

$$\mathcal{A}_i = \langle L_i, L_i(C_i) \rangle, \quad L_i \cdot C_i = n_i + b_i - 1,$$

then every $L\beta^ \mathcal{A}_i$ is b_i -twisted adherent to C'_i and contains $\mathcal{O}_{C'_i}(b_i - 1)$. In particular, untwisted adherence is preserved when all $b_i = 0$. These conclusions hold after any finite sequence of point blow-ups whose centers avoid the successive strict transforms of all the C_i , allowing infinitely near centers.*

Proof. The one-curve argument applies componentwise. Since β is an isomorphism near every C_i , for any integer b_i we have

$$L\beta^* \mathcal{O}_{C_i}(b_i - 1) \simeq \mathcal{O}_{C'_i}(b_i - 1), \quad \beta^* C_i = C'_i.$$

For the line bundles,

$$\beta^*(L_i(C_i)) = (\beta^* L_i)(C'_i), \quad (\beta^* L_i) \cdot C'_i = L_i \cdot C_i = n_i + b_i - 1.$$

A point blow-up preserves the cohomology of the structure sheaf, and derived pullback is fully faithful by Orlov's formula. The adherent presentations therefore persist. Induction proves the final assertion. Taking all $b_i = 0$ also proves the compatibility assertion. $\square$

This lemma preserves already-existing adherent blocks; it does not construct simultaneous adherent blocks. For one cyclic point, the following proposition reformulates the weighted-projective-plane model of [19, Example 3.17] in our notation.

Proposition 4.5 (KKS weighted-projective-plane model). *Let $n \geq 2$, and let $\tau : \mathbb{F}_n \rightarrow \mathbb{P}(1, 1, n)$ contract the negative section C_0 , with $C_0^2 = -n$. Writing f for the fiber class, the full exceptional collection*

$$\langle \mathcal{O}(-C_0 - f), \mathcal{O}(-f), \mathcal{O}, \mathcal{O}(C_0 + nf) \rangle$$

has an untwisted adherent block $\tilde{\mathcal{A}}_0 = \langle \mathcal{O}(-C_0 - f), \mathcal{O}(-f) \rangle$ over C_0 . It descends to

$$D^b(\text{coh } \mathbb{P}(1, 1, n)) = \langle D^b(K(n, 1)\text{-mod}), \mathcal{O}, \mathcal{O}(n) \rangle,$$

where the Kalck–Karmazyn algebra is

$$K(n, 1) \cong \mathbb{C}[z_1, \dots, z_{n-1}] / (z_1, \dots, z_{n-1})^2.$$

Proof. The collection is the projective-bundle collection used in [19, Example 3.17], with their exceptional section and fiber class denoted here by C_0 and f . Its fullness also follows from (1) with $a = n + 1$: twist by $\mathcal{O}(-f)$, then move the last term to the front by the Serre functor $(- \otimes \mathcal{O}(-2C_0 - (n + 2)f))[2]$ and omit the shift of that individual exceptional object. The divisor sequence

$$0 \longrightarrow \mathcal{O}(-C_0 - f) \longrightarrow \mathcal{O}(-f) \longrightarrow \mathcal{O}_{C_0}(-1) \longrightarrow 0$$

puts $\mathcal{O}_{C_0}(-1)$ in $\tilde{\mathcal{A}}_0$, proving compatibility. The equality $(-C_0 - f) \cdot C_0 = n - 1$ proves untwisted adherence. The surface is rational with rational singularities, so KKS descent [19, Theorem 2.12(i)] and identification [19, Theorem 3.16] apply. The remaining blocks descend equivalently to the line bundles $\mathcal{O}$ and $\mathcal{O}(n)$, since $\tau^* \mathcal{O}(n) = \mathcal{O}(C_0 + nf)$. [19, Example 3.14(2)] gives the displayed presentation of $K(n, 1)$. $\square$

The base decomposition is the existing KKS model. Its role here is to provide the initial adherent block for Lemma 4.4; the consequence for surfaces with a prescribed blow-up presentation is the following.

Corollary 4.6 (One-point descent under blow-ups). *Let X be a normal projective surface with a unique singularity of type $\frac{1}{n}(1, 1)$, where $n \geq 2$. Suppose its minimal resolution $\nu : S \rightarrow X$ is obtained from $\mathbb{F}_n$ by m point blow-ups whose centers avoid the negative section C_0 and its successive strict transforms, and that the exceptional locus of ν is the strict transform of C_0 . Then*

$$D^b(\text{coh } X) = \langle D^b(K(n, 1)\text{-mod}), F_1, \dots, F_{m+2} \rangle,$$

where all F_i are exceptional. The centers may be infinitely near, provided they lie over the smooth locus and avoid the contracted curve.

Proof. Start with the three blocks in [Proposition 4.5](#) on $\mathbb{F}_n$. At each blow-up β with exceptional curve D , the right-handed Orlov decomposition is

$$D^b(\mathrm{coh} S') = \langle L\beta^* D^b(\mathrm{coh} S), \mathcal{O}_D \rangle.$$

By [Lemma 4.4](#), the first block remains untwisted adherent to the strict transform of C_0 . The other two initial blocks and the m new blocks are exceptional; for infinitely near centers, their objects are pulled back in the derived sense from their creation stages. Full faithfulness preserves exceptionality. The decomposition is compatible with ν , whose only exceptional curve is assigned to the first block. By [Theorem 4.2](#), all other blocks descend equivalently, and [\[19, Theorem 3.16\]](#) identifies the first with $D^b(K(n, 1)\text{-mod})$. $\square$

This applies to the one-point Cavey–Prince cascades starting from $\mathbb{P}(1, 1, n)$ [\[6\]](#) whenever the resolution presentation satisfies these hypotheses. For $n = 3$, the starting model is $\langle D^b(K(3, 1)\text{-mod}), \mathcal{O}, \mathcal{O}(3) \rangle$, the specialization used in [Section 5](#).

For several singularities, finding the initial simultaneous adherent blocks is a separate problem. KKS gives a global obstruction in terms of the divisor class group.

Theorem 4.7 (KKS obstruction, [\[19, Corollary 4.8\]](#)). *Let X be a normal projective rational surface with cyclic quotient singularities, and let $\tau : Y \rightarrow X$ be its minimal resolution with exceptional divisor $\mathcal{D}$. Suppose there is a semiorthogonal decomposition*

$$D^b(\mathrm{coh} Y) = \langle \tilde{\mathcal{A}}_1, \dots, \tilde{\mathcal{A}}_t \rangle, \quad \mathcal{D} = \bigsqcup_{i=1}^t \mathcal{D}_i,$$

such that every nonempty $\mathcal{D}_i$ is a connected component of $\mathcal{D}$ and every $\tilde{\mathcal{A}}_i$ is untwisted adherent to $\mathcal{D}_i$ in the sense of [\[19, Definition 3.6\]](#). For $\mathcal{D}_i = \emptyset$, adherence means that $\tilde{\mathcal{A}}_i$ is generated by one line bundle, using the convention of [\[19, Remark 3.8\]](#). Then $\mathrm{Cl}(X)$ is torsion-free.

Proof. KKS [Corollary 3.18](#) explicitly allows empty chains; for such a chain the algebra is $\mathbb{C}$. Thus descent gives a decomposition into derived categories of local finite-dimensional algebras, including these singleton components. Additivity makes $G_0(X)$ free, and $G_0(X)_{\mathrm{tors}} \cong \mathrm{Cl}(X)_{\mathrm{tors}}$. This is the proof of [\[19, Corollary 4.8\]](#), using [\[19, Corollary 3.18, Lemma 4.1, and Proposition 4.4\]](#); the empty-chain convention is therefore included in the obstruction. $\square$

Bright’s exact sequence translates this obstruction into a Brauer-group calculation. We use the cohomological Brauer group $\mathrm{Br}(X) = H_{\mathrm{\acute{e}t}}^2(X, \mathbb{G}_m)_{\mathrm{tors}}$. In the present projective setting, its torsion classes are represented by Azumaya algebras by Gabber’s theorem, as used in [\[4, proof of Theorem 6\]](#); thus no distinction between the two Brauer groups arises for the finite groups computed here. This is also the Brauer group entering the obstruction in [\[19, Corollary 4.8\]](#).

Lemma 4.8 (Bright’s exact sequence [\[4, Proposition 1 and Section 3\]](#)). *Let X be a normal projective surface with rational singularities and rational minimal resolution $\tau : Y \rightarrow X$, and let $\Lambda_\tau \subset \mathrm{Pic}(Y)$ denote the subgroup generated by the exceptional curves. The intersection pairing induces a homomorphism*

$$\theta : \mathrm{Pic}(Y) \longrightarrow \Lambda_\tau^\vee, \quad D \longmapsto (E \mapsto D \cdot E),$$

and there is an exact sequence

$$0 \longrightarrow \mathrm{Pic}(X) \longrightarrow \mathrm{Pic}(Y) \xrightarrow{\theta} \Lambda_\tau^\vee \longrightarrow \mathrm{Br}(X) \longrightarrow \mathrm{Br}(Y).$$

Here $\mathrm{Br}(Y) = 0$, and the cokernel of θ is finite.

Corollary 4.9. *Let X be a normal projective surface with quotient singularities and rational minimal resolution $\tau : Y \rightarrow X$. For the intersection map of Lemma 4.8, there is an isomorphism*

$$\mathrm{Br}(X) \cong \mathrm{coker}(\theta : \mathrm{Pic}(Y) \longrightarrow \Lambda_\tau^\vee).$$

In particular, $\mathrm{Br}(X) = 0$ if and only if there are divisor classes $L_i \in \mathrm{Pic}(Y)$ with $L_i \cdot E_j = \delta_{ij}$ for all exceptional curves E_j .

Proof. Quotient singularities are rational, so Bright’s sequence applies, and $\mathrm{Br}(Y) = 0$. Surjectivity of θ is equivalent to its image containing every element of the basis dual to the exceptional curves. $\square$

All log del Pezzo surfaces in this paper satisfy the rationality hypothesis by Proposition 2.2. In this setting, the vanishing in Corollary 4.9 is equivalent to the KKS torsion-free condition: the Weil divisor class group $\mathrm{Cl}(X)$ has no torsion [19, Definition 4.7]. It is necessary for an untwisted adherent decomposition by [19, Corollary 4.8]; the absence of this obstruction alone does not construct such a decomposition.

For the Corti–Heuberger application, take the model $\mathbb{P}(1, 1, 3)$. The exceptional lattice of $\mathbb{F}_3 \rightarrow \mathbb{P}(1, 1, 3)$ is $\mathbb{Z}C_0$. The relation $f \cdot C_0 = 1$ makes the intersection map surjective, so Corollary 4.9 gives

$$\mathrm{Br}(\mathbb{P}(1, 1, 3)) = 0.$$

This calculation agrees with the $n = 3$ specialization of the KKS model in Proposition 4.5. Appendix B computes the Brauer obstruction throughout the six Corti–Heuberger cascades; unlike the one-point model, vanishing there does not by itself supply simultaneous adherent blocks.

5 Corti–Heuberger cascades and the surface X_{10}

The cascade classification supplies the base collections and the blow-up presentation needed for the Euler form and KKS descent of X_{10} . We begin with the deformation terminology that underlies this classification.

Definition 5.1 (qG -deformations and rigidity). Let X be a normal projective surface with quotient singularities. A $\mathbb{Q}$ -Gorenstein deformation, or qG -deformation, of X is a flat deformation that locally lifts to equivariant deformations of the index-one covers of the quotient singularities. The index-one cover is the local cyclic cover on which the canonical divisor becomes Cartier. For projective surfaces we use proper families, and qG -deformation equivalence means being connected by a chain of such families over connected bases [1, Basic Concepts, Definition 2 and proof of Lemma 6]. A surface is qG -rigid if every sufficiently small qG -deformation is trivial.

Rigidity of the surface is a global condition, distinct from rigidity of its singularities. The singularity $\frac{1}{3}(1, 1)$ is locally qG -rigid, so small qG -deformations preserve its analytic type; the degree is also locally constant [7, Section 1.1.1].

Corti–Heuberger classify the non-smooth del Pezzo surfaces with only $\frac{1}{3}(1, 1)$ -points into 29 families, grouped into six cascades [7, Theorems 3–4]. The operations relating these families are described by the following terminology.

Definition 5.2 (Floating (-1) -curves and cascades). Let X be a log del Pezzo surface with only $\frac{1}{3}(1, 1)$ -points. A *floating (-1) -curve* on X is a smooth rational curve contained in its smooth locus with self-intersection -1 [7, Definition 5]. A *cascade* connects the deformation families by blow-ups at nonsingular points, or equivalently by reversing contractions of floating (-1) -curves.

Contracting a floating (-1) -curve is the inverse of a blow-up at a nonsingular point and increases K^2 by one. These operations preserve the singular points and their analytic types.

Definition 5.3 (Marked blow-up presentations). A *marked blow-up presentation* of a smooth surface Y is a sequence of point blow-ups

$$Y_0 \xleftarrow{\beta_1} Y_1 \xleftarrow{\beta_2} \cdots \xleftarrow{\beta_m} Y_m \simeq Y,$$

with fixed identifications of the starting and final surfaces and, at each stage, a specified point $q_i \in Y_{i-1}$, its incidences with the strict transforms of previously introduced exceptional curves, and a label for the new exceptional divisor $D_i \subset Y_i$. For this general resolution presentation, the centers may be infinitely near points of Y_0 . When Y is a resolution of a surface with quotient singularities, the marking also specifies the curves contracted to those singularities. A *marked cascade presentation* specifies distinct smooth points on the chosen model of Z_k , their order, and the identification of their blow-up with the member X .

The distinctness in a cascade follows from ampleness. In any sequence of smooth-point blow-ups ending in a log del Pezzo surface X , an exceptional curve has anticanonical degree one when created. If $h \geq 1$ later centers lie on its successive strict transforms, its final strict transform D , still in the smooth locus, satisfies

$$(-K_X) \cdot D = 1 - h \leq 0, \quad (3)$$

contrary to ampleness. Thus the cascade centers are distinct on the base, and their exceptional curves are disjoint. This restriction does not apply to the internal resolution constructions of Appendix C.

We follow [7, Theorem 4]: $X_{k,d}$ is the index-one qG -deformation family with k singular points and $d = K_X^2$. The six bases occur in [7, Theorem 6(1)–(6)]. Their uniqueness is explicit in [7, Remark 7]: apart from the two auxiliary families, they are “the only isomorphism class of surfaces in that family”. We denote these surfaces by Z_k . Table 1 collects their invariants and the numerical data of the marked resolution models constructed in Appendix C. Here Y_0 is the starting smooth surface, $N_k = \text{rk } K_0(\tilde{Z}_k)$, and $\ell_k = N_k + k$ is the length of the resulting stack collection. The notation $S_{1,25/3}$ distinguishes $\mathbb{P}(1, 1, 3)$, which has Fano index five, from the index-one families $X_{k,d}$.

Proposition 5.4. *Each minimal resolution $\tilde{Z}_k$ has a marked presentation by successive point blow-ups of a smooth surface Y_0 , with infinitely near centers allowed. The starting surfaces and numbers of blow-ups are specified in Table 1. Appendix C records the incidence data, lists the centers, and constructs the resulting Orlov full exceptional collections. Applying the Ishii–Ueda embedding and adjoining the residual-gerbe objects gives full exceptional collections on the canonical stacks of the bases, of lengths ℓ_k as recorded in Table 1 and given by Corollary 3.7.*

k	Z_k	$K_{Z_k}^2$	Y_0	Number of blow-ups	N_k	ℓ_k
1	$S_{1,25/3} = \mathbb{P}(1, 1, 3)$	$25/3$	$\mathbb{F}_3$	0	4	5
2	$X_{2,17/3}$	$17/3$	$\mathbb{F}_3$	3	7	9
3	$X_{3,5}$	5	$\mathbb{F}_3$	4	8	11
4	$X_{4,7/3}$	$7/3$	$\mathbb{F}_2$	7	11	15
5	$X_{5,5/3}$	$5/3$	$\mathbb{P}^2$	9	12	17
6	$X_{6,2}$	2	$\mathbb{F}_0$	8	12	18

Table 1: Rigid cascade bases, marked resolution models, and collection lengths.

Appendix C proves Proposition 5.4 by constructing these presentations and collections.

Every surface in an index-one family $X_{k,d}$ is obtained from its listed base by $K_{Z_k}^2 - d$ blow-ups, respectively by [7, Corollary 8(1)–(6)], whose centers are “nonsingular points on” that base. Equation (3) also excludes infinitely near centers. The uniqueness of the bases is stronger than local rigidity of the singularities: it identifies any model in a base family with Z_k , allowing the computations in Appendix B to apply throughout each cascade. For the categorical construction, Proposition 5.4 fixes the initial collections. The remaining terms are the Orlov objects from the smooth blow-ups and the residual-gerbe objects over the stacky points; Corollary C.3 records the resulting order.

The two auxiliary families $B_{1,16/3}$ and $B_{2,8/3}$ have Fano index two and may also occur as endpoints of a directed MMP [7, Theorems 3 and 6]. By [7, Remark 9], a smooth-point blow-up of a member of either family admits an alternative sequence of four floating contractions to Z_1 , respectively Z_2 . Together with Z_1 , these are the three higher-index families in the classification.

The resulting stack collection on an index-one member $X \in X_{k,d}$ has length $12 - d + 4k/3$ by Corollary 3.7. We now make this construction explicit for degree-10 hypersurfaces $X_{10} \subset \mathbb{P}(1, 2, 3, 5)$, writing x_0, x_1, x_2, x_3 for coordinates of weights 1, 2, 3, 5. The description of the singularities of a general member is standard [18, 16]; to apply the construction to every quasismooth member, as required by Theorem 1.3, we record the following refinement.

Lemma 5.5. *A general degree-10 hypersurface in $\mathbb{P}(1, 2, 3, 5)$ is quasismooth. Every quasismooth member avoids the coordinate points of weights 2 and 5 and has a unique singular point $p = [0 : 0 : 1 : 0]$, of type $\frac{1}{3}(1, 1)$.*

Proof. For the memberwise assertion, quasismoothness forces the coefficients of x_1^5 and x_3^2 to be nonzero: these are the only degree-ten monomials whose first derivatives can be nonzero on the respective punctured coordinate axes. Otherwise the corresponding axis would lie in the cone and be singular. Thus the hypersurface avoids the ambient singular points of weights 2 and 5. At p , the quotient chart has weights $(1, 2, 5) \equiv (1, 2, 2) \pmod{3}$. Quasismoothness forces a nonzero coefficient of $x_0x_2^3$, the only monomial giving a linear term there. Eliminating x_0 leaves tangent weights $(2, 2)$, equivalent to $(1, 1)$. These are all the ambient singular points, so quasismoothness implies smoothness elsewhere. $\square$

For every member satisfying Lemma 5.5, weighted adjunction and the intersection formula of [10] give

$$\omega_{X_{10}}^\vee = \mathcal{O}_{X_{10}}(1), \quad K_{X_{10}}^2 = \frac{1}{3}.$$

The anticanonical class is ample, so X_{10} is a log del Pezzo surface. The higher-index families in [7, Theorem 3] have different numerical invariants, so the Fano index is one; [7, Theorem 4 and Table 2] identifies the family as $X_{1,1/3}$.

By [7, Corollary 8(1)], every such X_{10} is the blow-up of $\mathbb{P}(1, 1, 3)$ at eight smooth points, distinct by (3). Its minimal resolution is therefore the blow-up of $\mathbb{F}_3$ at eight distinct points off C_0 . This proves the memberwise geometric assertion of Theorem 1.3. The following lemma and proposition compare general centers and general hypersurfaces; they are not needed for this assertion. The ring presentation is due to Reid–Suzuki [24, Section 1 and Table 1.1].

Lemma 5.6 (Reid–Suzuki). *For eight distinct general smooth points $P_1, \dots, P_8 \in \mathbb{P}(1, 1, 3)$, their blow-up $\eta : S \rightarrow \mathbb{P}(1, 1, 3)$ is a log del Pezzo surface in $X_{1,1/3}$. Its anticanonical divisor and ring are*

$$-K_S = \eta^*(-K_{\mathbb{P}(1,1,3)}) - \sum_{i=1}^8 D_i,$$

$$R(S, -K_S) \cong \frac{\mathbb{C}[x, y, z, w]}{(f_{10})}, \quad \deg(x, y, z, w) = (1, 2, 3, 5),$$

where D_i is the exceptional curve over P_i and f_{10} is weighted homogeneous of degree 10. The surface S is isomorphic to its anticanonical model, a quasismooth hypersurface $X_{10} \subset \mathbb{P}(1, 2, 3, 5)$.

Proof. For ampleness, the preceding identification of the hypersurface family and [7, Corollary 8(1)] show that some member of $X_{1,1/3}$ is obtained by eight smooth-point blow-ups of $\mathbb{P}(1, 1, 3)$, with distinct centers by (3). The space of ordered distinct smooth eight-tuples is irreducible, and ampleness of the Cartier divisor $-3K$ is open in the corresponding family of blow-ups [25, Tag 0D2S]. Hence $-K_S$ is ample for general centers. The blow-up formula gives the asserted divisor and $K_S^2 = \frac{1}{3}$. The blow-ups preserve the unique $\frac{1}{3}(1, 1)$ -point and introduce no other singularities. Moreover, $(-K_S) \cdot D_i = 1$ forces Fano index one, so S belongs to $X_{1,1/3}$.

Since $-K_S$ is ample, the Proj of the Reid–Suzuki ring is S itself. The resulting hypersurface is quasismooth: away from the vertex, the anticanonical cone is locally a $\mathbb{G}_m$ -bundle over the smooth locus. Over the $\frac{1}{3}(1, 1)$ -point it is locally the quotient of $\mathbb{A}^2 \times \mathbb{G}_m$ by μ_3 , acting faithfully on the second factor through the canonical character. This action is free, so the punctured cone is smooth there as well. $\square$

The generality comparison strengthens the memberwise presentation as follows.

Proposition 5.7 (The general hypersurface). *A general quasismooth $X_{10} \subset \mathbb{P}(1, 2, 3, 5)$ is isomorphic to the anticanonical model of the blow-up of $\mathbb{P}(1, 1, 3)$ at eight distinct general smooth points. Its minimal resolution is therefore the blow-up of $\mathbb{F}_3$ at eight distinct points away from the negative section C_0 .*

In Appendix A, we prove the parameter counts and finiteness of blow-up presentations before combining them in the dominance argument.

With the blow-up presentation established, we can construct the collection and compute its Euler form. Fix any X_{10} satisfying the hypotheses of Theorem 1.3, and let

$$\sigma : \tilde{X}_{10} \longrightarrow \mathbb{F}_3$$

be a memberwise resolution presentation as above. Write $D_1, \dots, D_8$ for its pairwise disjoint exceptional curves and f for the fiber class on $\mathbb{F}_3$. Let

$$\nu : \tilde{X}_{10} \longrightarrow X_{10}$$

be the minimal resolution, and let E be the strict transform of C_0 . Before constructing the collection, the same geometry gives the Brauer-group vanishing relevant to descent:

$$\mathrm{Br}(X_{10}) = 0.$$

Indeed, E is the only curve contracted by ν , and

$$(\sigma^* f) \cdot E = f \cdot C_0 = 1.$$

This class makes the intersection map in [Lemma 4.8](#) surjective. Since $\tilde{X}_{10}$ is rational, Bright's exact sequence gives the claimed vanishing.

Orlov's formula applied to the standard collection on $\mathbb{F}_3$, followed by [Corollary 3.7](#), gives the collection displayed in [Theorem 1.3](#). The curves D_i are pairwise disjoint, so their exceptional objects may be listed in any order.

Corollary 5.8. *Let $\pi : \mathcal{X}_{10} \rightarrow X_{10}$ be the canonical stack. The category $D^b(\mathrm{coh} \mathcal{X}_{10})$ has a full exceptional collection of length 13.*

More generally, take $a = s + 4$ in the Hille–Perling family (1) [[15](#), Proposition 5.2]. This gives the following collections; $s = -1$ is the standard one.

Proposition 5.9 (The Euler form for $\mathcal{X}_{10}$). *Let $s \in \mathbb{Z}$ and let*

$$\begin{aligned} \mathcal{C}(s) = \langle \mathcal{E}_p, \Phi(\mathcal{O}_{D_1}(-1)), \dots, \Phi(\mathcal{O}_{D_8}(-1)), \Phi(\sigma^* \mathcal{O}), \Phi(\sigma^* \mathcal{O}(f)), \\ \Phi(\sigma^* \mathcal{O}(C_0 + (s+4)f)), \Phi(\sigma^* \mathcal{O}(C_0 + (s+5)f)) \rangle \end{aligned}$$

be an exceptional collection of $D^b(\mathrm{coh} \mathcal{X}_{10})$ associated with the Hille–Perling toric system for $\mathbb{F}_3$. The matrix of the Euler form $S_{ij}(s) = \chi(\mathcal{C}_i(s), \mathcal{C}_j(s))$ is

$$S_{\mathcal{X}_{10}}(s) = \left(\begin{array}{c|c|cccc} 1 & 0_{1 \times 8} & 0 & -1 & -(s+1) & -(s+2) \\ \hline 0_{8 \times 1} & I_8 & & & & -\mathbf{1}_{8 \times 4} \\ \hline & & 1 & 2 & 2s+7 & 2s+9 \\ & & 0 & 1 & 2s+5 & 2s+7 \\ & 0_{4 \times 1} & 0 & 0 & 1 & 2 \\ & & 0 & 0 & 0 & 1 \end{array} \right).$$

Proof. This is the specialization $n = 3$, $m = 8$ of [Proposition 6.3](#). The curves D_i are disjoint, so their objects may be ordered arbitrarily. The four line bundles have degrees $0, 1, s+1, s+2$ on E , giving the displayed first row. $\square$

The KKS descent framework of [Section 4](#) applies to the singular coarse surface X_{10} .

Theorem 5.10. *Every X_{10} satisfying the hypotheses of [Theorem 1.3](#) admits a semiorthogonal decomposition*

$$D^b(\mathrm{coh} X_{10}) = \langle D^b(K(3, 1)\text{-mod}), F_1, \dots, F_{10} \rangle,$$

where $\eta : X_{10} \rightarrow \mathbb{P}(1, 1, 3)$ is the blow-down of the chosen presentation, ν is the minimal resolution, and

$$\begin{aligned} F_1 &= \mathcal{O}_{X_{10}}, & F_2 &= \eta^* \mathcal{O}_{\mathbb{P}(1,1,3)}(3), \\ F_{i+2} &= \mathcal{O}_{\nu(D_i)} & (1 \leq i \leq 8) \end{aligned}$$

are exceptional objects.

This is [Corollary 4.6](#) with $n = 3$ and $m = 8$, using the memberwise presentation above. Since the centers are distinct and ν is an isomorphism near every D_i , the right-handed Orlov objects descend to $\mathcal{O}_{\nu(D_i)}$. If $\tau : \mathbb{F}_3 \rightarrow \mathbb{P}(1, 1, 3)$ contracts C_0 , then $\tau\sigma = \eta\nu$ and $\tau^* \mathcal{O}_{\mathbb{P}(1,1,3)}(3) = \mathcal{O}_{\mathbb{F}_3}(C_0 + 3f)$. The projection formula and $R\nu_* \mathcal{O} = \mathcal{O}_{X_{10}}$ therefore identify the two descended line bundles with F_1, F_2 .

Corollary 5.11. *Let X belong to an index-one family $X_{1,d}$. Choose a marked presentation by blowing up m distinct smooth points of $\mathbb{P}(1, 1, 3)$, so that $m = 25/3 - d$. Then*

$$D^b(\text{coh } X) = \langle D^b(K(3, 1)\text{-mod}), F_1, \dots, F_{m+2} \rangle,$$

where $F_1, \dots, F_{m+2}$ are exceptional objects.

Proof. The marked presentation induces the resolution presentation required by [Corollary 4.6](#), with $n = 3$; its centers are distinct on $\mathbb{F}_3$ and avoid the negative section by (3). Apply that corollary. $\square$

6 Euler matrices for canonical-stack collections

[Corollary 3.6](#), proved in [Section 3](#), supplies the stack collections considered here. We first give the mixed-basket Euler block formula, separating the local blocks from the resolution matrix and the pairings between them. For line-bundle entries on the resolution, those pairings depend only on intersections with the exceptional curves. We then specialize to a single cyclic point with a Hirzebruch-surface blow-up presentation and compute every block.

Proposition 6.1 (Mixed-basket Euler block formula). *Let X be a normal projective rational surface with singular points $p_1, \dots, p_r$ of types $\frac{1}{n_i}(1, 1)$, where $n_i \geq 2$, and let $E_i \subset \tilde{X}$ be the exceptional curve over p_i . Fix a full exceptional collection $\langle G_1, \dots, G_N \rangle$ on the minimal resolution. For the associated stack collection of [Corollary 3.6](#), the Euler matrix is*

$$S = \begin{pmatrix} T_{n_1} & 0 & \cdots & 0 & B_1 \\ 0 & T_{n_2} & \cdots & 0 & B_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & T_{n_r} & B_r \\ 0 & 0 & \cdots & 0 & H \end{pmatrix}, \quad H_{ab} = \chi(G_a, G_b).$$

For an arbitrary resolution collection, the local-to-resolution blocks are defined by $(B_i)_{j,a} = \chi(\mathcal{E}_{p_i,j}, \Phi(G_a))$, where $2 \leq j \leq n_i - 1$; no line-bundle hypothesis is imposed at this stage. The local block is

$$T_n = (I_{n-2} - J_{n-2})^2, \quad (J_{n-2})_{ab} = \delta_{b,a+1} \quad (1 \leq a, b \leq n-2), \quad (4)$$

where J_{n-2} is the nilpotent shift with ones immediately above the diagonal. For $n = 2$, the block is empty.

If, in addition, $G_a = \mathcal{O}(A_a)$ are line bundles, these pairings are determined by intersections. Put $d_{ia} = A_a \cdot E_i$. For $n_i \geq 4$,

$$(B_i)_{j,a} = \begin{cases} 0, & 2 \leq j \leq n_i - 3, \\ d_{ia} + 1, & j = n_i - 2, \\ -d_{ia}, & j = n_i - 1. \end{cases}$$

For $n_i = 3$, the sole row is $(B_i)_{2,a} = -d_{ia}$; for $n_i = 2$, B_i has no rows. Thus the resolution matrix H and the intersection numbers $A_a \cdot E_i$ determine the entire Euler matrix for a line-bundle collection on the resolution.

Proof. The equivariant Koszul resolution on $[\mathbb{A}^2/\mu_n]$ gives

$$\chi(\mathcal{O}_0 \otimes \rho_u, \mathcal{O}_0 \otimes \rho_v) = \delta_{uv} - 2\delta_{v,u+1} + \delta_{v,u+2}.$$

Within $2 \leq u, v \leq n-1$, no cyclic wraparound term occurs, so this is T_n . Distinct local blocks have disjoint support, which gives the off-diagonal zeros among them. Full faithfulness of Φ gives H , and [Theorem 3.4](#) supplies the lower-left zeros.

Let Ψ be the left adjoint of Φ . The calculation (2) applies near each E_i . For the indices belonging to the non-special block, it gives

$$\Psi(\mathcal{E}_{p_i, n_i-2}) = \mathcal{O}_{E_i}(-n_i)[1], \quad \Psi(\mathcal{E}_{p_i, n_i-1}) = \mathcal{O}_{E_i}(-n_i + 1),$$

with all earlier non-special characters sent to zero. The first displayed source belongs to the local block only when $n_i \geq 4$, and the second only when $n_i \geq 3$. For the divisor inclusion $E_i \hookrightarrow \tilde{X}$, duality and $E_i^2 = -n_i$ give

$$\chi_{\tilde{X}}(\mathcal{O}_{E_i}(t), \mathcal{O}(A_a)) = -\chi_{\mathbb{P}^1}(\mathcal{O}(d_{ia} - n_i - t)).$$

Adjunction of Ψ and Φ , together with $\chi(\mathcal{O}_{\mathbb{P}^1}(q)) = q + 1$, gives $d_{ia} + 1$ for the shifted penultimate image and $-d_{ia}$ for the final image. These are precisely the claimed rows. $\square$

Remark 6.2 (Local checks and orthogonality). For a vector bundle V on $[\mathbb{A}^2/\mu_n]$, gerbe adjunction and the ring-action convention of [Section 3](#) give

$$\mathrm{RHom}(\mathcal{O}_0 \otimes \rho_j, V) \simeq (V_0 \otimes \rho_{-j-2})^{\mu_n}[-2].$$

Its Euler characteristic is the multiplicity of ρ_{j+2} in the fiber V_0 . In particular, $\chi(\mathcal{E}_j, \mathcal{L}_a) = \delta_{a, j+2}$, with indices read modulo n . For $n \geq 3$ and the resolution generator $P_1 = p^*\mathcal{O}_{\mathbb{P}^1}(1)$, the identification in [Section 3](#) gives

$$0 \longrightarrow \phi(P_1) \longrightarrow \mathcal{L}_{n-1} \longrightarrow \mathcal{E}_{n-1} \longrightarrow 0.$$

Hence $\chi(\mathcal{E}_{n-1}, \phi(P_1)) = -1$. This directly checks the coupling to $\mathcal{O}(f)$, including the -1 in the X_{10} matrix. The fiber-multiplicity rule cannot be applied to $\phi(P_1)$ itself, since it is not locally free.

For $n \geq 5$, the objects $\mathcal{E}_j$ with $2 \leq j \leq n-3$ are orthogonal to the resolution image on both sides: one direction is semiorthogonality, and the other follows from $\psi(\mathcal{E}_j) = 0$. This does not split off an orthogonal summand of the whole category, since $\chi(\mathcal{E}_j, \mathcal{E}_{j+1}) = -2$ and $\chi(\mathcal{E}_j, \mathcal{E}_{j+2}) = 1$.

The local categorical structure is independent of the global configuration. What remains geometry-dependent is the resolution block and, for line-bundle entries, their intersections with the exceptional curves. We specialize [Proposition 6.1](#) to a single $\frac{1}{n}(1, 1)$ -point whose resolution is obtained from $\mathbb{F}_n$ by m point blow-ups away from the successive strict transforms of C_0 .

To describe the collection and its matrix blocks, let $n \geq 2$ and $m \geq 0$. Suppose that a normal projective surface X has a minimal resolution fitting into a diagram

$$\tilde{X} \xrightarrow{\sigma} \mathbb{F}_n, \quad \tilde{X} \xrightarrow{\nu} X,$$

where σ is a sequence of m point blow-ups, possibly infinitely near, whose centers avoid the successive strict transforms of the negative section $C_0 \subset \mathbb{F}_n$, and ν contracts the strict transform E of C_0 to the unique singular point of X , of type $\frac{1}{n}(1, 1)$. Let D_a be the exceptional curve at the a -th blow-up and G_a the derived pullback of $\mathcal{O}_{D_a}(-1)$ through the subsequent blow-ups. We use the intersection conventions

$$C_0^2 = -n, \quad C_0 \cdot f = 1, \quad f^2 = 0.$$

Write $\mathcal{E}_i = \mathcal{E}_{p,i}$ for the objects of [Definition 3.3](#) at the unique singular point, and retain the notation $\mathcal{X}, \Phi$ of [Proposition 6.1](#). For $s \in \mathbb{Z}$, the projective-bundle collection (1) with $a = s + n + 1$, Orlov's blow-up formula, and the Ishii–Ueda decomposition give the ordered full exceptional collection

$$\begin{aligned} \mathcal{C}_{n,m}(s) = \langle & \mathcal{E}_2, \dots, \mathcal{E}_{n-1}, \Phi(G_m), \dots, \Phi(G_1), \\ & \Phi(\sigma^*\mathcal{O}), \Phi(\sigma^*\mathcal{O}(f)), \Phi(\sigma^*\mathcal{O}(C_0 + (s + n + 1)f)), \\ & \Phi(\sigma^*\mathcal{O}(C_0 + (s + n + 2)f)) \rangle. \end{aligned}$$

The local block is T_n from (4). For $n \geq 4$, the local-to-Hirzebruch block $B_n(s) \in \text{Mat}_{(n-2) \times 4}(\mathbb{Z})$ is

$$B_n(s) = \begin{pmatrix} & & 0_{(n-4) \times 4} & \\ 1 & 2 & s+2 & s+3 \\ 0 & -1 & -(s+1) & -(s+2) \end{pmatrix}, \quad (5)$$

where the zero block is absent for $n = 4$. The penultimate and final rows are indexed by $\mathcal{E}_{n-2}$ and $\mathcal{E}_{n-1}$, respectively. For $n = 3$, the local block consists only of $\mathcal{E}_2$, so $T_3 = (1)$. The representation $\rho_{n-2} = \rho_1$ is special and does not belong to the Ishii–Ueda complement; consequently, only the final row of $B_n(s)$ occurs:

$$B_3(s) = \begin{pmatrix} 0 & -1 & -(s+1) & -(s+2) \end{pmatrix}.$$

For $n = 2$, both T_2 and $B_2(s)$ are empty, and the collection consists only of the resolution objects.

The Hirzebruch-surface block $H_n(s) \in M_4(\mathbb{Z})$ is

$$H_n(s) = \begin{pmatrix} 1 & 2 & 2s+n+4 & 2s+n+6 \\ 0 & 1 & 2s+n+2 & 2s+n+4 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Proposition 6.3 (The one-point Hirzebruch-surface specialization). *For the collection $\mathcal{C}_{n,m}(s)$ and the blocks defined above, the $(n+m+2) \times (n+m+2)$ matrix of the Euler form is*

$$S_{n,m}(s) = \begin{pmatrix} T_n & 0 & B_n(s) \\ 0 & I_m & -\mathbf{1}_{m \times 4} \\ 0 & 0 & H_n(s) \end{pmatrix}.$$

Here $\mathbf{1}_{m \times 4}$ denotes the matrix all of whose entries are equal to 1.

Proof. Proposition 6.1 gives T_n and the local pairings with the line bundles. Since σ is an isomorphism near E , $(\sigma^*M) \cdot E = M \cdot C_0$ for a divisor class M on $\mathbb{F}_n$. For the four line bundles in $\mathcal{C}_{n,m}(s)$, the intersection numbers are

$$M \cdot C_0 = 0, \quad 1, \quad s+1, \quad s+2.$$

These values give (5), including the stated truncation when $n = 3$. The adjoint images in (2) are supported on E . Since every G_a is supported away from E , the remaining local pairings satisfy

$$\chi(\mathcal{E}_i, \Phi(G_a)) = 0$$

for every i and a .

For a blow-up $\beta : S' \rightarrow S$ at p , with exceptional curve D , divisor duality gives, for any perfect complex A ,

$$\mathrm{RHom}_{S'}(\mathcal{O}_D(-1), L\beta^*A) \simeq Li_p^*A[-1], \quad \chi(\mathcal{O}_D(-1), L\beta^*A) = -\mathrm{rk}(A).$$

Indeed, the normal bundle $\mathcal{O}_D(-1)$ in the exceptional inverse image cancels the source twist, leaving the constant complex Li_p^*A on D and a shift by -1 . Earlier Orlov objects have rank zero, and line bundles have rank one. Induction, full faithfulness, and Orlov semiorthogonality [23, Theorem 4.3 and Corollary 4.4] therefore give

$$\chi(G_a, G_b) = \delta_{ab}, \quad \chi(G_a, \sigma^*L) = -1, \quad \chi(\sigma^*L, G_a) = 0$$

for every line bundle L on $\mathbb{F}_n$. These pairings yield I_m , $-\mathbf{1}_{m \times 4}$, and the opposite zero block, even for infinitely near centers. In that case the off-diagonal Ext groups need not vanish, but their Euler characteristics still do.

The last block follows from Riemann–Roch on $\mathbb{F}_n$. Using

$$K_{\mathbb{F}_n} = -2C_0 - (n+2)f$$

together with

$$\chi(\mathcal{O}(A), \mathcal{O}(B)) = 1 + \frac{1}{2}(B - A) \cdot (B - A - K_{\mathbb{F}_n}),$$

gives $H_n(s)$ for the four Hille–Perling line bundles and completes the specialization of Proposition 6.1. $\square$

For $n = 3$ and $m = 8$, one has $T_3 = (1)$ and $B_3(s) = (0, -1, -(s+1), -(s+2))$, while $H_3(s)$ is the 4×4 block in Proposition 5.9. Together with I_8 and $-\mathbf{1}_{8 \times 4}$, the formula recovers $S_{\mathcal{X}_{10}}(s)$.

In particular, Proposition 6.3 applies to the one-point Cavey–Prince cascades starting from $\mathbb{P}(1, 1, n)$ whenever their resolution presentations satisfy the stated hypotheses [6], including infinitely near centers.

Concluding remarks

The coarse-surface constructions now cover the one- and three-point cascades, by [Corollaries B.6](#) and [5.11](#). For the six-point cascade, [Corollary B.8](#) gives a decomposition after a nonzero Brauer twist; untwisted adherent descent remains obstructed. Here k counts $\frac{1}{3}(1, 1)$ -points. For $k = 2, 4, 5$, constructing simultaneous untwisted adherent blocks remains open despite Brauer vanishing. Separately, comparisons with specific equation parameter spaces, as in [Appendix A](#), are discussed in [Remark C.4](#).

For other mixed baskets, [Proposition 6.1](#) reduces the Euler calculation for a line-bundle resolution collection to its resolution matrix and intersection data. The Cayey–Prince models $X_{n_1+n_2} \subset \mathbb{P}(1, 1, n_1, n_2)$ [[6](#), [Corollary 7.1](#)] provide a natural testing ground. Mutation equivalence of collections arising from different presentations is another question.

A The general degree-ten hypersurface as a blow-up

The memberwise presentation is already proved in [Section 5](#). Here we prove the generality comparison of [Proposition 5.7](#). The forward construction in [Lemma 5.6](#) associates a quasismooth degree-10 hypersurface to eight general smooth points of $\mathbb{P}(1, 1, 3)$. To compare generality in these two parameter spaces, we show that configurations in any prescribed dense open locus account for a dense open subset of hypersurfaces. The lemmas below separate the coordinate changes, parameter counts, and finite ambiguity of the blow-up presentation before establishing this dominance.

Throughout this appendix, put

$$\begin{aligned} U &\subset \mathbb{P}H^0(\mathbb{P}(1, 2, 3, 5), \mathcal{O}(10)), & G &= \operatorname{Aut} \mathbb{P}(1, 2, 3, 5), \\ Q &= \mathbb{P}(1, 1, 3), & H &= \operatorname{Aut} Q, \end{aligned}$$

where U is the nonempty open set of quasismooth hypersurfaces. Let W be a nonempty H -invariant open subset of the space of ordered distinct smooth eight-tuples on Q for which [Lemma 5.6](#) holds. For $P = (P_1, \dots, P_8) \in W$, write $S_P = \operatorname{Bl}_{P_1, \dots, P_8} Q$. Thus $-K_{S_P}$ is ample, $K_{S_P}^2 = 1/3$, and S_P is its own anticanonical model.

Lemma A.1 (Lifting graded isomorphisms). *Let $S = \mathbb{C}[x, y, z, w]$ with weights $(1, 2, 3, 5)$, and let $f, g \in S$ be nonzero forms of degree 10. Every graded isomorphism $S/(f) \simeq S/(g)$ lifts uniquely to a graded automorphism of S .*

Proof. There are no relations in degrees below 10. The images of the four generators therefore have unique polynomial lifts in their respective degrees. Since these degrees are distinct, the lifts have the form

$$\begin{aligned} x &\longmapsto a_1x, \\ y &\longmapsto a_2y + bx^2, \\ z &\longmapsto a_3z + cxy + dx^3, \\ w &\longmapsto a_5w + q_5(x, y, z), \end{aligned}$$

where q_5 is weighted homogeneous of degree 5. The induced isomorphism on the space of indecomposables S_+/S_+^2 forces each a_i to be nonzero. The substitution is consequently triangular and can be inverted successively in the order x, y, z, w . It is a graded polynomial automorphism inducing the prescribed quotient isomorphism. Uniqueness follows from uniqueness of the four lifts. $\square$

Lemma A.2 (Hypersurface parameters). *The space U is irreducible of dimension 19, and G has dimension 11. Two hypersurfaces in U are isomorphic if and only if they lie in the same G -orbit. The stabilizer of a hypersurface in G identifies with its automorphism group.*

Proof. In coordinates of weights 1, 2, 3, 5, a degree-10 monomial $x^a y^b z^c w^e$ satisfies $a + 2b + 3c + 5e = 10$. For fixed e, c , there are $\lfloor (10 - 5e - 3c)/2 \rfloor + 1$ choices of b , with a then determined. The contributions for $e = 0, 1, 2$ are 14, 5, 1, respectively. Hence there are 20 monomials, and U is a nonempty open subset of $\mathbb{P}^{19}$.

The group G consists of graded coordinate changes modulo weighted scalar multiplication [8, Theorem 4.2]. The numbers of monomials in the four generator degrees 1, 2, 3, 5 are 1, 2, 3, 6, so after quotienting by weighted scalars, $\dim G = 11$. For clarity, the degree-5 monomials are $w, x^5, x^3y, xy^2, x^2z, yz$; requiring the diagonal coefficients of a graded change to be nonzero is an open condition and does not alter the dimension.

Adjunction gives $\mathcal{O}_X(1) \simeq \mathcal{O}_X(-K_X)$ as reflexive sheaves for $X \in U$. An isomorphism of hypersurfaces thus induces a graded isomorphism of their anticanonical section rings. By Lemma A.1, it extends to a graded coordinate change, proving the assertion about orbits. The extension is unique modulo weighted scalars: changing the identification of the anticanonical sheaf multiplies degree- d sections by the d -th power of a scalar, and the polynomial lifts are otherwise unique. Consequently restriction identifies the stabilizer in G with $\text{Aut}(X)$. $\square$

Lemma A.3 (Configurations and their stabilizers). *The space W is irreducible of dimension 16, and H has dimension 8. There is a nonempty open subset of W on which the stabilizer of every ordered tuple in H is trivial.*

Proof. The configuration space is an open subset of $(Q_{\text{sm}})^8$, so it is irreducible of dimension 16. In coordinates u, v, z of weights 1, 1, 3, the graded automorphisms of Q have the form

$$(u, v) \mapsto A(u, v), \quad z \mapsto az + q_3(u, v),$$

modulo weighted scalars, with $A \in \text{GL}_2$, $a \neq 0$, and q_3 a binary cubic [8, Theorem 4.2]. Thus $\dim H = 8$.

To compute a general stabilizer, first require that the tuple have at least three distinct images in $\mathbb{P}_{[u:v]}^1$. An automorphism fixing the ordered tuple then induces the identity on this $\mathbb{P}^1$. On the chart $u \neq 0$, with $t = v/u$ and $\zeta = z/u^3$, the remaining action is

$$(t, \zeta) \mapsto (t, a\zeta + q(t)), \quad \deg q \leq 3.$$

Writing $q(t) = c_0 + c_1t + c_2t^2 + c_3t^3$, the fixing equations are

$$(a - 1)\zeta_j + c_0 + c_1t_j + c_2t_j^2 + c_3t_j^3 = 0, \quad 1 \leq j \leq 8.$$

Their coefficient matrix has rows $(\zeta_j, 1, t_j, t_j^2, t_j^3)$. To show that a maximal minor is not identically zero, evaluate it at any configuration with five distinct t_j and $\zeta_j = t_j^4$ for those five points. At such a configuration, the minor is, up to sign, the Vandermonde determinant

$$\det(t_j^{\ell-1})_{1 \leq j, \ell \leq 5} = \prod_{1 \leq i < j \leq 5} (t_j - t_i) \neq 0.$$

Hence the rank-five locus is a nonempty open subset of the configuration chart and meets W . The equations $\zeta_j = t_j^4$ serve only to exhibit one configuration in this locus; they impose no condition on the general tuple. On the rank-five locus, $a = 1$ and $q = 0$. Together with the identity action on $\mathbb{P}^1$, this proves that the stabilizer is trivial. $\square$

The next lemma controls the information lost when the ordered centers are forgotten. Its finiteness assertion is what allows the configuration count to be used in the dominance argument.

Lemma A.4 (Finiteness of blow-up presentations). *For $P \in W$, the surface S_P has finitely many floating (-1) -curves. A fixed isomorphism class of surfaces S_P arises from only finitely many H -orbits in W . If the stabilizer of P in H is trivial, then $\text{Aut}(S_P)$ is finite.*

Proof. Let $\nu : Y \rightarrow S_P$ be the minimal resolution and put $A = \nu^*(-K_{S_P})$. Then $A^2 = 1/3$. A floating (-1) -curve is disjoint from the singular locus, so its strict transform C on Y satisfies

$$C^2 = -1, \quad A \cdot C = 1.$$

Decompose its numerical class as $C = 3A + v$. Then

$$A \cdot v = 0, \quad v^2 = (C - 3A)^2 = C^2 - 6A \cdot C + 9A^2 = -4.$$

The Hodge index theorem makes $A^\perp$ negative definite [14, Chapter V, Theorem 1.9]. The possible vectors v lie in a translate of the numerical divisor lattice and on the compact locus $v^2 = -4$ in this negative definite space. There are therefore only finitely many possible integral classes C . Each contains at most one irreducible curve: two distinct curves in that class would have intersection -1 , whereas distinct irreducible curves on a smooth surface have nonnegative intersection. This proves finiteness of the floating curves.

Any presentation of a fixed surface S_P as the blow-up of eight distinct smooth points of Q selects an ordered set of eight disjoint floating curves to contract. There are only finitely many such ordered sets. For each set, the contraction determines the target and the ordered centers up to an isomorphism of the target with Q , that is, up to H . This proves the second assertion.

Finally, $\text{Aut}(S_P)$ permutes the finite set of floating curves. A finite-index subgroup fixes each of the eight curves of the chosen presentation and descends to an automorphism of Q fixing the ordered tuple P . This descent is injective, since an automorphism inducing the identity on Q is already the identity away from the exceptional curves, a dense open subset. If the stabilizer of P is trivial, the finite-index subgroup is trivial, so $\text{Aut}(S_P)$ is finite. $\square$

We can now prove dominance by computing the fibers of a family of equations. This makes the subtraction of the eight-dimensional group H explicit, without requiring a global moduli space of surfaces.

Proposition A.5 (Dominance of the blow-up construction). *The locus in U of hypersurfaces isomorphic to S_P for some $P \in W$ contains a dense open subset of U . The same conclusion holds after replacing W by any nonempty open subset of it.*

Proof. Choose a nonempty open subset $W_0 \subset W$ on which the tuple stabilizers are trivial, as in Lemma A.3. We first explain how to choose hypersurface equations algebraically on a further nonempty open subset of W_0 . Blowing up the eight tautological sections of $Q \times W_0$ gives an algebraic family with fibers S_P . By Lemma 5.6, each anticanonical ring has generators of degrees 1, 2, 3, 5 and a single relation of degree 10. For every

$m \geq 0$, Kawamata–Viehweg vanishing for klt surfaces [20, Theorem 2.70], applied to the reflexive sheaf $\mathcal{O}_{S_P}(-mK_{S_P})$, gives

$$H^1(S_P, \mathcal{O}_{S_P}(-mK_{S_P})) = 0,$$

since $-(m+1)K_{S_P}$ is ample. The blow-ups leave the family unchanged from $Q \times W_0$ near its singular locus, and the family is smooth elsewhere, so these relative reflexive powers are flat and restrict to the stated fiber sheaves. Cohomology and base change, after shrinking W_0 , therefore make the section spaces and multiplication maps in degrees at most 10 vector bundles and bundle maps compatible with the fibers. Choose local sections representing the four generators. The degree-10 relations then form a line bundle, which can also be trivialized after shrinking. A generator of this line supplies a degree-10 equation whose coefficients vary regularly with P . Thus we obtain a morphism

$$h : W_0 \longrightarrow U, \quad V(h(P)) \simeq S_P,$$

where $V(h(P))$ denotes the hypersurface defined by the projective form $h(P)$.

Allowing all graded coordinate changes gives the morphism

$$F : G \times W_0 \longrightarrow U, \quad (g, P) \longmapsto g \cdot h(P).$$

Its image consists precisely of the equations of hypersurfaces isomorphic to some S_P , $P \in W_0$, by Lemma A.2. We compute the dimension of each nonempty fiber of F .

Fix $u \in F(G \times W_0)$. Projection of $F^{-1}(u)$ to W_0 has image

$$\{P \in W_0 : S_P \simeq V(u)\}.$$

By Lemma A.4, this is a finite union of intersections of H -orbits with W_0 . Every nonempty such intersection has dimension 8: the tuple stabilizer is trivial, $\dim H = 8$, and W_0 is open. Here we use the orbit-dimension formula [5, Proposition 1.11]. For each tuple in this image, the choices of g satisfying $g \cdot h(P) = u$ form a coset of the stabilizer of $h(P)$. This stabilizer is finite by Lemmas A.2 and A.4. The projection therefore has finite fibers, and $\dim F^{-1}(u) = 8$.

The fiber-dimension theorem now gives

$$\dim F(G \times W_0) = \dim G + \dim W_0 - 8 = \dim U.$$

Since U is irreducible, the image is dense. It is constructible by Chevalley’s theorem [25, Tag 054K], and hence contains a dense open subset of U [25, Tag 005K]. Starting with any prescribed nonempty open subset of W leaves all of the preceding arguments and dimensions unchanged. This proves the final assertion as well. $\square$

Proof of Proposition 5.7. By Proposition A.5, a general hypersurface in U is isomorphic to S_P with eight distinct general smooth centers. Since $-K_{S_P}$ is ample, S_P is its anticanonical model. The minimal resolution $\tau : \mathbb{F}_3 \rightarrow Q$ is an isomorphism over the smooth locus of Q . The eight centers therefore lift uniquely to eight distinct points on $\mathbb{F}_3$ away from C_0 . Blowing up these points resolves S_P ; its exceptional locus over the singular point is still the single (-3) -curve C_0 , so this resolution is minimal. $\square$

B Brauer groups and obstructions in the Corti–Heuberger cascades

We prove [Theorem 1.2](#): the Brauer group vanishes in the cascades with $1 \leq k \leq 5$ singular points and is $\mathbb{Z}/3$ in the six-point cascade. The proof combines the cascade description of [Section 5](#), the cokernel criterion of [Corollary 4.9](#), and invariance under smooth-point blow-ups.

We first describe the lattice mechanism for a normal projective rational surface X with a finite set of singularities p_i of types $\frac{1}{n_i}(1, 1)$, where $n_i \geq 2$. Its minimal resolution has pairwise disjoint exceptional curves $E_1, \dots, E_r$ with $E_i^2 = -n_i$. Put $\Lambda = \bigoplus_i \mathbb{Z}E_i$. In the notation of [Lemma 4.8](#), the intersection map is

$$\theta : \text{Pic}(\tilde{X}) \longrightarrow \Lambda^\vee, \quad D \longmapsto (D \cdot E_1, \dots, D \cdot E_r).$$

Here $\Lambda^\vee = \text{Hom}(\Lambda, \mathbb{Z})$ has basis $e_1, \dots, e_r$ dual to the E_i ; in particular, $\theta(E_i) = -n_i e_i$. By [Corollary 4.9](#), $\text{Br}(X) \cong \text{coker}(\theta)$.

By uniqueness of the bases in [Section 5](#), a toric or k -gon model in a base family is the base itself. The four-point model used below is instead in $X_{4,4/3}$; its comparison with $Z_4 = X_{4,7/3}$ requires one smooth-point blow-up. No deformation-invariance assertion for the Brauer group is needed in these comparisons.

Bright’s exact sequence applies to arbitrary rational surface singularities [[4](#), Proposition 1], whereas the classification carried out in [[4](#), Section 3] concerns the classical singular del Pezzo case obtained by contracting (-2) -curves. In the Corti–Heuberger cascades the exceptional curves over the singular points have self-intersection -3 , so the calculations below are not contained in that classification.

B.1 Glue in the exceptional lattice

The intersection lattice $P = \text{Pic}(\tilde{X})$ is free and unimodular. Indeed, the rational-surface description used in [Corollary 2.6](#) reduces this to $\mathbb{P}^2$ and the Hirzebruch surfaces, whose intersection matrices have determinant ± 1 . Each point blow-up adds an orthogonal rank-one summand of square -1 . In particular, intersection identifies P with $P^\vee$.

Definition B.1. The *saturation* of the exceptional lattice in P is

$$\Lambda^{\text{sat}} = (\Lambda \otimes \mathbb{Q}) \cap \text{Pic}(\tilde{X}).$$

The lattice is *primitive* if P/Λ is torsion-free, equivalently if $\Lambda^{\text{sat}} = \Lambda$. The finite quotient $\Lambda^{\text{sat}}/\Lambda$ is its *glue group*. Using the intersection form to identify $\Lambda^\vee$ with a lattice in $\Lambda \otimes \mathbb{Q}$, the *discriminant group*

$$A_\Lambda := \Lambda^\vee / \Lambda \cong \bigoplus_{i=1}^r \mathbb{Z}/n_i$$

carries the induced discriminant bilinear form

$$b_\Lambda : A_\Lambda \times A_\Lambda \longrightarrow \mathbb{Q}/\mathbb{Z}, \quad b_\Lambda(\bar{x}, \bar{y}) = x \cdot y \pmod{\mathbb{Z}}.$$

An *integral overlattice* of Λ is a finite-index extension inside $\Lambda \otimes \mathbb{Q}$ whose pairings are integral. It lies in $\Lambda^\vee$, and its quotient by Λ is totally isotropic for b_Λ , meaning that every pairing between elements of the quotient is zero. Conversely, the inverse image of

such a subgroup is an integral overlattice. This is the bilinear version of the overlattice correspondence discussed in [21, Section 1.4]; the quadratic-form version for even lattices is not needed here. In particular, $\Lambda^{\text{sat}}/\Lambda$ is totally isotropic, since its representatives belong to the integral lattice P .

The next lemma identifies the Brauer group canonically with the dual of the glue group, making the failure of primitivity an explicit obstruction.

Lemma B.2 (The Brauer group and the glue group). *There are natural isomorphisms*

$$\text{Br}(X) \cong \text{coker}(\theta) \cong \text{Hom}(\Lambda^{\text{sat}}/\Lambda, \mathbb{Q}/\mathbb{Z}).$$

Thus $\text{Br}(X) = 0$ if and only if Λ is primitive in P . As abstract finite abelian groups, $\text{Br}(X)$ and $\Lambda^{\text{sat}}/\Lambda$ are isomorphic, but this last identification is not canonical.

Proof. Applying $\text{Hom}(-, \mathbb{Z})$ to $0 \rightarrow \Lambda \rightarrow P \rightarrow P/\Lambda \rightarrow 0$ gives an exact sequence

$$P^\vee \rightarrow \Lambda^\vee \rightarrow \text{Ext}^1(P/\Lambda, \mathbb{Z}) \rightarrow 0,$$

because P is free. Under the unimodular identification $P \cong P^\vee$, the first map is θ . For any finitely generated abelian group G , the sequence $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$ gives $\text{Ext}^1(G, \mathbb{Z}) \cong \text{Hom}(G_{\text{tors}}, \mathbb{Q}/\mathbb{Z})$. Here $(P/\Lambda)_{\text{tors}} = \Lambda^{\text{sat}}/\Lambda$. The result follows from Corollary 4.9; a finite abelian group is noncanonically isomorphic to its character group. $\square$

The Gram matrix $\text{diag}(-n_1, \dots, -n_r)$ gives $\Lambda^\vee = \bigoplus_i \mathbb{Z}(E_i/n_i)$. Every glue class thus has a representative

$$L = \sum_{i=1}^r \frac{a_i}{n_i} E_i \in \text{Pic}(\tilde{X}), \quad \bar{a}_i \in \mathbb{Z}/n_i,$$

where the a_i are integer lifts of the residue classes. Since the intersection form is integral and adjunction gives $K_{\tilde{X}} \cdot E_i = n_i - 2$, every such representative satisfies

$$L^2 = - \sum_i \frac{a_i^2}{n_i} \in \mathbb{Z}, \quad K_{\tilde{X}} \cdot L = \sum_i \frac{a_i(n_i - 2)}{n_i} \in \mathbb{Z}. \quad (6)$$

These are necessary numerical constraints on glue classes, not a sufficiency criterion for their realization in $\text{Pic}(\tilde{X})$.

For the remainder of this appendix, specialize to k points all of type $\frac{1}{3}(1, 1)$, so $r = k$, all $n_i = 3$, and $A_\Lambda \cong (\mathbb{Z}/3)^k$. Corti–Heuberger’s defect is $\sigma = \dim_{\mathbb{F}_3}(\Lambda^{\text{sat}}/\Lambda)$. They determine it in [7, Section 2.1, Proposition 18(b), and Lemma 28]: $\sigma = 1$ for $X_{6,2}, X_{6,1}$ and $\sigma = 0$ otherwise. These values are stated in Section 2.1, although omitted from their tables. Together with Lemma B.2, they already give the group orders in Theorem 1.2. The calculations below exhibit the intersection maps and glue classes realizing these values. The next lemma specializes (6) and explains why the one- and two-point cases need no choice of model.

Lemma B.3. *Every class in $\Lambda^{\text{sat}}/\Lambda$ can be represented in the form*

$$L = \frac{1}{3} \sum_{i=1}^k a_i E_i, \quad a_i \in \{0, \pm 1\}.$$

Such a representative satisfies the congruences

$$\sum_{i=1}^k a_i^2 \equiv 0 \pmod{3}, \quad \sum_{i=1}^k a_i \equiv 0 \pmod{3}.$$

Consequently:

- (i) if $k = 1$ or $k = 2$, the lattice Λ is primitive;
- (ii) if $3 \leq k \leq 5$, every nonzero glue class is represented by

$$\pm \frac{E_a + E_b + E_c}{3}$$

for a triple $\{a, b, c\}$, and the glue subgroup $\Lambda^{\text{sat}}/\Lambda$ has order at most 3;

- (iii) if $k = 6$, classes supported on three or on six exceptional curves are numerically possible. The glue subgroup actually realized on Z_6 is determined in [Section B.4](#).

Proof. For $n_i = 3$, choose the residue representatives $a_i \in \{0, \pm 1\}$. Formula (6) becomes

$$-\frac{1}{3} \sum_i a_i^2 \in \mathbb{Z}, \quad \frac{1}{3} \sum_i a_i \in \mathbb{Z},$$

which gives the two stated congruences.

The first congruence says that the number of nonzero coefficients is divisible by 3. This proves (i). If $3 \leq k \leq 5$, a nonzero class has support of size 3, and the second congruence forces its three coefficients to have the same sign. By the cited overlattice correspondence, the glue subgroup is totally isotropic for b_Λ . After changing their signs, two support-three classes with distinct supports A and B have pairing

$$-\frac{\#(A \cap B)}{3} \pmod{\mathbb{Z}}.$$

For two distinct triples in a set of at most five elements, the intersection has size 1 or 2, so the two classes cannot be orthogonal. Hence all nonzero glue classes generate the same cyclic subgroup, proving (ii). For $k = 6$, supports of size 3 and 6 satisfy the first numerical condition, subject also to the stated congruence on the signs. In particular, the alternating coefficient vector satisfies both congruences. Thus the numerical conditions permit the coefficient pattern $\frac{1}{3}(E_1 - E_2 + E_3 - E_4 + E_5 - E_6)$, as asserted in (iii). $\square$

Corollary B.4. *Let X be a log del Pezzo surface with at most two singular points, all of type $\frac{1}{3}(1, 1)$. Its Brauer group is trivial. In particular, $\text{Br}(Z_1) = \text{Br}(\mathbb{P}(1, 1, 3)) = 0$ and $\text{Br}(Z_2) = 0$.*

Proof. Combine [Lemmas B.2](#) and [B.3](#); the smooth case is the rational-surface vanishing used in [Lemma 4.8](#). $\square$

The bound of two singularities in [Corollary B.4](#) uses the (-3) -lattice congruences; no analogous conclusion for arbitrary n_i is asserted here. For $k = 1$, this recovers the direct fiber-class computation in [Lemma 4.8](#). For $3 \leq k \leq 5$, the same lemmas leave only the possibilities 0 and $\mathbb{Z}/3$. The explicit models below decide between them; the numerical congruences alone do not prove that an allowed glue class occurs.

B.2 Behavior under cascade blow-ups

The morphisms in a Corti–Heuberger cascade are blow-ups of nonsingular points by [7, Corollary 8]. Their effect on the intersection map is elementary.

Lemma B.5. *Let X be a normal projective surface with quotient singularities and rational minimal resolution, and let $X' \rightarrow X$ be the blow-up of a nonsingular point. This blow-up preserves the Brauer group:*

$$\mathrm{Br}(X') \cong \mathrm{Br}(X).$$

The same conclusion holds for a finite sequence of smooth-point blow-ups, including infinitely near centers.

Proof. On minimal resolutions, the induced morphism

$$\beta : \tilde{X}' \longrightarrow \tilde{X}$$

is the blow-up of a point disjoint from the exceptional curves $E_1, \dots, E_k$ of $\tilde{X} \rightarrow X$, since the resolution is an isomorphism over the smooth locus. The strict transforms of these curves give natural identifications

$$\Lambda_{X'} \cong \Lambda_X, \quad \Lambda_{X'}^\vee \cong \Lambda_X^\vee.$$

If D is the new exceptional curve, the Picard group decomposes as

$$\mathrm{Pic}(\tilde{X}') = \beta^* \mathrm{Pic}(\tilde{X}) \oplus \mathbb{Z}D, \quad D \cdot E_i = 0.$$

This decomposition implies that the intersection numbers satisfy

$$(\beta^* L + nD) \cdot E_i = L \cdot E_i$$

for every $L \in \mathrm{Pic}(\tilde{X})$ and every i . Under the displayed identification of the dual exceptional lattices, the image and cokernel of θ are therefore unchanged. The Brauer groups are these cokernels by Corollary 4.9; naturality of Bright’s sequence identifies this isomorphism with pullback. The assertion for a sequence follows step by step, even when a center lies on a curve introduced by an earlier blow-up: that curve lies over the smooth locus, not over a singular point of the current surface. $\square$

Combining this lemma with the smooth-point presentations supplied by [7, Corollary 8] gives, for every $X \in X_{k,d}$,

$$\mathrm{Br}(X) \cong \mathrm{Br}(Z_k). \tag{7}$$

It is therefore enough to compute the cokernel on one model in each base family, or on a surface connected to that base by smooth-point blow-ups. Uniqueness of the base, rather than a choice of a general member, justifies this reduction.

B.3 The cases $k = 3, 4, 5$

Three points: a toric model. The three-point base admits a direct toric calculation.

Corti–Heuberger exhibit polygon 20 as a toric qG -degeneration in the family $X_{3,5}$ [7, Section 7, Table 4 and Figure 11]. We check its singularities directly before using uniqueness of the base. Let $N \cong \mathbb{Z}^2$, and let $v_1, \dots, v_5 \in N$ be the primitive ray

generators of its fan, in counterclockwise order, with V_i the corresponding torus-invariant divisors:

$$v_1 = (1, 0), \quad v_2 = (-1, 3), \quad v_3 = (-2, 3), \quad v_4 = (-1, 0), \quad v_5 = (0, -1).$$

The first three two-dimensional cones have determinant 3, while the remaining two are smooth. The primitive rays inserted to resolve the three singular cones are

$$u_{12} = (0, 1), \quad u_{23} = (-1, 2), \quad u_{34} = (-1, 1).$$

Let E_1, E_2, E_3 be the corresponding exceptional torus-invariant curves. Each inserted ray is one third of the sum of its original neighbors, and the determinants of all adjacent rays in the subdivided fan equal one. Thus the subdivision is the minimal toric resolution, obtained by the ray-subdivision procedure of [9, Chapters 10–11]. Computing the boundary self-intersections from the fan relations as in [9, Theorem 10.4.4] gives the following table; the last column records $w_{i-1} + w_{i+1} = b_i w_i$, with square $-b_i$ for the corresponding curve:

curve	square	ray relation
V_1	0	$v_5 + u_{12} = 0$
E_1	-3	$v_1 + v_2 = 3u_{12}$
V_2	-1	$u_{12} + u_{23} = v_2$
E_2	-3	$v_2 + v_3 = 3u_{23}$
V_3	-1	$u_{23} + u_{34} = v_3$
E_3	-3	$v_3 + v_4 = 3u_{34}$
V_4	-1	$u_{34} + v_5 = v_4$
V_5	0	$v_4 + v_1 = 0$

The inserted curves are therefore precisely the exceptional (-3) -curves, so the surface has exactly three $\frac{1}{3}(1, 1)$ -points and no other singularities. After the unimodular change of coordinates $(x, y) \mapsto (x, x + y)$, these are the rays of the polygon cited above. It belongs to $X_{3,5}$ (no higher-index family has three singular points) and hence is isomorphic to Z_3 by the uniqueness statement recalled in Section 5.

The boundary intersections give

$$\theta(V_1) = e_1, \quad \theta(V_2) = e_1 + e_2, \quad \theta(V_4) = e_3.$$

Consequently, the dual basis vectors satisfy

$$e_1 = \theta(V_1), \quad e_2 = \theta(V_2) - \theta(V_1), \quad e_3 = \theta(V_4),$$

so θ is surjective and the Brauer group vanishes:

$$\mathrm{Br}(Z_3) = \mathrm{Br}(X_{3,5}) = 0.$$

Equivalently, the divisor classes

$$\mathcal{O}(-V_1), \quad \mathcal{O}(V_1 - V_2), \quad \mathcal{O}(-V_4)$$

have intersection $-\delta_{ij}$ with E_j . This differs from the dual-basis convention in Corollary 4.9 only by replacing every L_i by $-L_i$; the same sign convention is used in the six-point obstruction.

Corollary B.6 (KKS decomposition in the three-point cascade). *Let $X \in X_{3,d}$ and $s = 5 - d$. Then*

$$D^b(\mathrm{coh} X) = \left\langle D^b(K(3,1)\text{-mod}), D^b(K(3,1)\text{-mod}), D^b(K(3,1)\text{-mod}), F_1, \dots, F_{2+s} \right\rangle,$$

where every F_i is exceptional.

Proof. The toric base Z_3 has torsion-free class group. Of its five fixed points, the first three in the displayed fan are quotient singularities and the last two are smooth. KKS [19, Proposition 5.6, Remark 5.8, and Corollary 5.10] therefore gives three untwisted adherent blocks on the resolution and two singleton blocks. Here M in KKS Remark 5.8 is the correcting line bundle on $\tilde{Z}_3$ used in the tensor twist $M \otimes \mathcal{O}(K_{\tilde{Z}_3})$: its degree is zero on every exceptional curve except the final curve over the last fixed point, where it is -1 . Since that fixed point is smooth, the exception is absent and one takes $M = \mathcal{O}$. By [7, Corollary 8(3)], X is obtained by s smooth-point blow-ups of Z_3 , at distinct centers by (3). On resolutions these avoid all three (-3) -curves. Lemma 4.4 preserves the three adherent blocks, and the right-handed Orlov formula adds s exceptional objects at the right. KKS descent and identification [19, Theorems 2.12(i) and 3.16] give the stated decomposition. $\square$

Remark B.7 (Toric surfaces and toric degenerations). Among nonsmooth del Pezzo surfaces with only $\frac{1}{3}(1,1)$ -points, the toric surfaces are precisely those in

$$S_{1,25/3}, \quad X_{1,22/3}, \quad X_{1,19/3}, \quad X_{3,5}, \quad X_{3,4}, \quad X_{6,2}.$$

These are the toric cases in the Corti–Heuberger classification [7, Tables 1, 2, and 4]; see also [22, Table 1]. In particular, Z_k is toric precisely for $k \in \{1, 3, 6\}$.

The toric degenerations of Z_2, Z_4, Z_5 in [7, Table 4, polygons 23, 11, and 7] have additional T -singularities: respectively an A_1 -point, an A_2 -point, and two $\frac{1}{4}(1,1)$ -points. Thus these toric special fibers are not the rigid bases themselves. The toric construction of [19, Section 5] applies to the special fibers, but does not directly give decompositions of the non-toric bases Z_2, Z_4, Z_5 ; such decompositions do not follow merely from the existence of a qG -degeneration.

The k -gon construction. For the four- and five-point calculations we use the k -gon construction. It starts with a smooth del Pezzo surface of degree k containing a cyclic configuration $C_1 + \dots + C_k$ of (-1) -curves, blows up the vertices $C_i \cap C_{i+1}$, and contracts the strict transforms of the C_i . Corti–Heuberger show that the resulting surface has k singularities of type $\frac{1}{3}(1,1)$, degree $k/3$, and belongs to the family $X_{k,k/3}$ [7, Section 3.5, “ k -gons”]. Thus the models used below are respectively members of $X_{4,4/3}$ and $X_{5,5/3}$. Let D_i be the exceptional curve over the i -th vertex and E_i the strict transform of C_i , with cyclic indices. Direct intersection calculations give $E_i^2 = -3$, since each C_i passes through two vertices. The curves E_i are pairwise disjoint, and each D_i meets just E_i and E_{i+1} , transversally once. Consequently,

$$\theta(D_i) = e_i + e_{i+1}, \quad \theta(E_i) = -3e_i.$$

The degree also follows from the discrepancy computation in Corollary 3.7: the k blow-ups lower K^2 from k to zero, and contraction of the k disjoint (-3) -curves raises it to $k/3$.

Five points. For the five-point base, consider the degree-five del Pezzo surface $\text{Bl}_{p_1, \dots, p_4} \mathbb{P}^2$, with exceptional classes Q_i and line class H . The curves

$$Q_1, \quad H - Q_1 - Q_2, \quad Q_2, \quad H - Q_2 - Q_3, \quad H - Q_1 - Q_4$$

have adjacent intersections one and nonadjacent intersections zero, so they form a cyclic pentagon of (-1) -curves. The construction above gives a member of $X_{5,5/3}$, which is the rigid base Z_5 by uniqueness. The five vectors $e_i + e_{i+1}$ are the columns of

$$A_5 = I + P_5,$$

where P_5 is the cyclic permutation matrix. A direct calculation gives

$$\det(A_5) = \prod_{\zeta^5=1} (1 + \zeta) = 2.$$

Consequently, $\mathbb{Z}^5/A_5\mathbb{Z}^5$ has order 2. Since the image of θ also contains $3\mathbb{Z}^5$, and multiplication by 3 is invertible on this order-two quotient, it follows that

$$A_5\mathbb{Z}^5 + 3\mathbb{Z}^5 = \mathbb{Z}^5.$$

It follows that θ is surjective and the Brauer group vanishes:

$$\text{Br}(Z_5) = \text{Br}(X_{5,5/3}) = 0.$$

Four points. For the four-point cascade, it is convenient to use a member of $X_{4,4/3}$, obtained from the rigid base $Z_4 = X_{4,7/3}$ by one blow-up at a nonsingular point [7, Corollary 8]. By Lemma B.5, it has the same cokernel as Z_4 . The vectors $e_i + e_{i+1}$ span a rank-three subgroup, so one additional divisor class is needed. To avoid confusing the exceptional classes on a degree-four del Pezzo surface with the (-3) -curves above, write

$$S = \text{Bl}_{p_1, \dots, p_5} \mathbb{P}^2$$

with basis $H, Q_1, \dots, Q_5$, where $H^2 = 1$, $Q_i^2 = -1$, and all other pairings between basis elements vanish. Consider

$$\begin{aligned} C_1 &= Q_1, \\ C_2 &= H - Q_1 - Q_2, \\ C_3 &= Q_2, \\ C_4 &= 2H - Q_1 - Q_2 - Q_3 - Q_4 - Q_5. \end{aligned}$$

Their self-intersections are all -1 . The adjacent intersections are

$$C_1 \cdot C_2 = C_2 \cdot C_3 = C_3 \cdot C_4 = C_4 \cdot C_1 = 1,$$

and the nonadjacent intersections are

$$C_1 \cdot C_3 = 0, \quad C_2 \cdot C_4 = 0.$$

Their intersection graph is therefore a 4-cycle. In addition,

$$Q_3 \cdot C_1 = Q_3 \cdot C_2 = Q_3 \cdot C_3 = 0, \quad Q_3 \cdot C_4 = 1.$$

For general $p_1, \dots, p_5$, these intersections are transverse; in particular, Q_3 avoids the four vertices of the cycle. If π is their blow-up, the projection formula gives

$$\theta(\pi^*Q_3) = e_4.$$

Thus, we obtain

$$\begin{aligned} e_1 &= \theta(D_4) - e_4, \\ e_2 &= \theta(D_1) - e_1, \\ e_3 &= \theta(D_2) - e_2. \end{aligned}$$

It follows that θ is surjective and the 4-gon representative has trivial Brauer group. Since it is a one-point blow-up of Z_4 , [Lemma B.5](#) gives

$$\mathrm{Br}(Z_4) = \mathrm{Br}(X_{4,7/3}) \cong \mathrm{Br}(X_{4,4/3}) = 0.$$

B.4 The six-point obstruction

Unlike the five smaller cascades, the six-point cascade carries a nonzero Brauer obstruction. We first exhibit a relation among the toric boundary divisors that produces an order-three glue class in $\Lambda^{\mathrm{sat}}/\Lambda \subset A_\Lambda$. This proves nonvanishing, but does not yet determine the whole group. We then compute the full image of the intersection map: a single alternating-sum congruence modulo three describes it, showing that the Brauer group is exactly $\mathbb{Z}/3$, with no additional independent obstruction.

For the rigid base $Z_6 = X_{6,2}$ of [Section 5](#), Corti–Heuberger give the corresponding toric model in [[7](#), Section 2.2.5]. We compute the intersection map by first describing the minimal toric resolution and then determining the lattice generated by the images of its boundary divisors. Let $N \cong \mathbb{Z}^2$, and let $v_1, \dots, v_6 \in N$ be the primitive ray generators of its fan, with V_i the corresponding torus-invariant divisors:

$$\begin{aligned} v_1 &= (1, 0), & v_2 &= (-1, 3), & v_3 &= (-2, 3), \\ v_4 &= (-1, 0), & v_5 &= (1, -3), & v_6 &= (2, -3), \end{aligned}$$

in counterclockwise order. The unimodular change $(x, y) \mapsto (x, x+y)$ sends these vertices to those of [[7](#), Figure 1]. Indices are cyclic modulo 6 throughout. Every adjacent cone has determinant 3, and hence defines a cyclic quotient singularity of order 3. The primitive rays inserted in the minimal toric resolution are

$$u_i = \frac{v_i + v_{i+1}}{3}, \quad i \in \mathbb{Z}/6\mathbb{Z}.$$

The inserted rays are explicitly

$$\begin{aligned} u_1 &= (0, 1), & u_2 &= (-1, 2), & u_3 &= (-1, 1), \\ u_4 &= (0, -1), & u_5 &= (1, -2), & u_6 &= (1, -1). \end{aligned}$$

Let E_i be the exceptional torus-invariant curve corresponding to u_i . Since

$$\det(v_i, u_i) = \det(u_i, v_{i+1}) = 1,$$

the subdivision is smooth and gives the minimal toric resolution [[9](#), Chapters 10–11]. The resolved boundary is the alternating cycle

$$V_1, E_1, V_2, E_2, \dots, V_6, E_6,$$

and the ray relations are

$$u_{i-1} + u_i = v_i, \quad v_i + v_{i+1} = 3u_i.$$

Therefore, [9, Theorem 10.4.4] gives

$$V_i^2 = -1, \quad E_i^2 = -3.$$

Consequently, the minimal resolution of each singularity consists of one (-3) -curve, so all six singularities are of type $\frac{1}{3}(1, 1)$.

A nonzero glue class. The obstruction comes from a divisibility relation. The principal-divisor formula $\text{div}(\chi^m) = \sum_\rho \langle m, v_\rho \rangle D_\rho$ of [9, Theorem 4.1.3], applied to the character $m = (0, 1)$, gives

$$\begin{aligned} 0 &= E_1 + 3V_2 + 2E_2 + 3V_3 + E_3 - E_4 \\ &\quad - 3V_5 - 2E_5 - 3V_6 - E_6. \end{aligned}$$

Rearranging gives

$$E_1 - E_2 + E_3 - E_4 + E_5 - E_6 = 3L, \tag{8}$$

where the divisor L is

$$L = -V_2 - V_3 + V_5 + V_6 - E_2 + E_5 \in \text{Pic}(\tilde{Z}_6).$$

Thus L lies in Λ^{sat} but not in Λ : the E_i are linearly independent, and its coefficients in this basis are $\pm 1/3$. Its class has order three and realizes the alternating pattern allowed by Lemma B.3. Moreover, $L^2 = -2$, confirming the isotropy of its class for the discriminant pairing in Definition B.1. In particular, Lemma B.2 already implies that $\text{Br}(Z_6) \neq 0$. We next determine the whole group.

The cokernel. For a complete toric variety, the torus-invariant prime divisors generate $\text{Cl}(\tilde{Z}_6)$, and smoothness gives $\text{Cl}(\tilde{Z}_6) = \text{Pic}(\tilde{Z}_6)$ [9, Theorem 4.1.3]. Hence the toric boundary generates $\text{Pic}(\tilde{Z}_6)$, and its image under θ is generated by

$$\theta(E_i) = -3e_i, \quad \theta(V_j) = e_{j-1} + e_j,$$

with cyclic indices. Put $c_j = e_{j-1} + e_j$, where $e_0 = e_6$, and let $A \subset \mathbb{Z}^6$ be the subgroup generated by $c_2, \dots, c_6$. These five vectors, together with e_6 , form a basis of $\mathbb{Z}^6$: the matrix of their columns is lower triangular with diagonal entries one. Thus A is primitive of rank five. Moreover,

$$c_1 = c_2 - c_3 + c_4 - c_5 + c_6,$$

so A is also the span of all six images $\theta(V_j)$. Consider the surjective homomorphism

$$\lambda : \mathbb{Z}^6 \longrightarrow \mathbb{Z}, \quad (a_1, \dots, a_6) \longmapsto a_1 - a_2 + a_3 - a_4 + a_5 - a_6.$$

Every c_j lies in $\ker(\lambda)$. Since both A and $\ker(\lambda)$ have rank five and A is primitive, this inclusion is an equality. The image of the full toric boundary is therefore

$$\text{im}(\theta) = A + 3\mathbb{Z}^6 = \ker(\lambda) + 3\mathbb{Z}^6.$$

This sum is the kernel of the reduction of λ modulo 3. Indeed, if $\lambda(a) = 3q$, then $a - 3qe_1 \in \ker(\lambda)$, while the reverse inclusion is immediate. Consequently,

$$\mathrm{im}(\theta) = \left\{ (a_1, \dots, a_6) \in \mathbb{Z}^6 : a_1 - a_2 + a_3 - a_4 + a_5 - a_6 \equiv 0 \pmod{3} \right\}.$$

The homomorphism $\lambda \bmod 3 : \mathbb{Z}^6 \rightarrow \mathbb{Z}/3$ is surjective, so the cokernel and Brauer group are

$$\mathrm{coker}(\theta) \cong \mathbb{Z}/3, \quad \mathrm{Br}(Z_6) = \mathrm{Br}(X_{6,2}) \cong \mathbb{Z}/3.$$

By Lemma B.2, the glue group also has order three. It is therefore generated by the class of L in (8); no support-three glue class occurs on Z_6 . Indeed, its two nonzero elements are $[L]$ and $-[L]$, whose coefficient vectors modulo three are $\pm(1, -1, 1, -1, 1, -1)$; both have all six entries nonzero. Here the global toric divisor relations select the alternating support-six class, whereas the computations for $k = 3, 4, 5$ exclude even the numerically permitted support-three classes.

The obstruction to untwisted adherence. In particular, there cannot exist divisor classes L_i satisfying

$$L_i \cdot E_j = -\delta_{ij},$$

since such classes would make θ surjective.

By Theorem 4.7, an untwisted adherent decomposition would force $\mathrm{Cl}(X_{6,2})$ to be torsion-free, equivalently $\mathrm{Br}(X_{6,2}) = 0$. The computed nonzero class rules this out. The toric KKS construction instead gives a decomposition of a twisted category, and the twist propagates through the cascade.

Corollary B.8 (Twisted KKS decomposition in the six-point cascade). *Let $X \in X_{6,d}$, let $\eta : X \rightarrow Z_6$ be a cascade presentation, and put $s = 2 - d$. There is a generator $\alpha \in \mathrm{Br}(Z_6)$ such that, with $\alpha_X = \eta^* \alpha$,*

$$D^b(\mathrm{coh}(X, \alpha_X)) = \left\langle \underbrace{D^b(K(3,1)\text{-mod}), \dots, D^b(K(3,1)\text{-mod})}_{6 \text{ components}}, F_1, \dots, F_s \right\rangle.$$

Here $\mathrm{coh}(X, \alpha_X)$ uses the KKS convention of coherent right modules over an Azumaya algebra representing α_X , and the F_i are exceptional objects of its derived category; their list is empty when $s = 0$. The class α_X is nonzero.

Proof. Write $\pi : \tilde{Z}_6 \rightarrow Z_6$ for the resolution and use the counterclockwise boundary order $V_1, E_1, \dots, V_6, E_6$ above. In KKS notation, $E_{i,0} = V_i$, $E_{i,1} = E_i$, $m_i = 1$, and $d_{i,1} = -E_i^2 = 3$; the last fixed point corresponds to the cone between v_6 and v_1 . [19, Proposition 5.6, equations (5.4)–(5.6), p. 510] gives the block $\langle \mathcal{L}_{i,0}, \mathcal{L}_{i,0}(E_i) \rangle$, where

$$\mathcal{L}_{i,0} = \mathcal{O}\left(\sum_{j < i} (V_j + E_j) + V_i\right), \quad \deg(\mathcal{L}_{i,0}|_{E_i}) = 1 + \delta_{i6}.$$

Indeed, V_i meets E_i once, and the sum contributes only $V_1 \cdot E_6 = 1$ when $i = 6$. Our adherence convention $\deg(\mathcal{L}_{i,0}|_{E_i}) = 3 + b_i - 1$ therefore gives $b_i = -1 + \delta_{i6}$, exactly [19, Proposition 5.6, equation (5.7)]. Adjunction gives $K_{\tilde{Z}_6} \cdot E_i = 1$, so tensoring by $\mathcal{O}(K_{\tilde{Z}_6})$ adds one to every twist:

$$b = (-1, -1, -1, -1, -1, 0), \quad b + \theta(K_{\tilde{Z}_6}) = e_6.$$

We also fix the Brauer sign: KKS's map B sends the degree vector of a splitting bundle $\mathcal{V}$, with $\pi^*\mathcal{R} \cong \text{End}(\mathcal{V})$, to $[\mathcal{R}]$, not $[\mathcal{R}^{\text{op}}]$. The twisted pushforward is $F \mapsto R\pi_*(F \otimes \mathcal{V}^\vee)$, a right $\mathcal{R}$ -module, as in [19, Proposition 4.9 and equations (4.13)–(4.14), pp. 498–500]. Since $B \circ \theta = 0$, the displayed equality gives $\alpha = B(b) = B(e_6)$, agreeing with [19, equation (5.8) and Theorem 5.9]. Finally $\lambda(e_6) = -1$, so this class generates $\text{Br}(Z_6)$. There are no smooth fixed points, so the base decomposition has six $K(3, 1)$ -components and no singleton blocks.

On the resolution of X , Lemma 4.4 preserves these twists. The right-handed Orlov formula adds s singleton blocks. The intersection-map description in Lemma B.5 identifies the resulting Brauer class with $\eta^*\alpha$, still a generator. More explicitly, a vector bundle on $\tilde{Z}_6$ realizing the twists, as in [19, Corollary 4.10(1)], pulls back to one with the same restrictions on the six exceptional curves. Use the twisted pushforward of [19, equation (4.14)]. The componentwise argument of [19, Theorem 4.19] identifies the six adherent components with $D^b(K(3, 1)\text{-mod})$. For the additional Orlov objects, the restrictions to the six exceptional curves are zero, so [19, Lemma 4.14] identifies each with the twisted pullback of its pushforward. Full faithfulness of twisted pullback preserves their exceptionality; adjunction preserves their semiorthogonality to the preceding components. Fullness follows from [19, Lemma 4.13] and the full resolution decomposition. Thus the singleton blocks require no line-bundle adherence hypothesis. This is twisted descent, not ordinary pushforward under Theorem 4.2. $\square$

B.5 Proof of Theorem 1.2

We first collect the computations on the bases. For $k = 1, 2$, Corollary B.4 gives vanishing independently of the model. For $k = 3, 5$, Section B.3 computes the Brauer group on a toric surface and a pentagon surface in the respective base families. Each is the corresponding Z_k , by uniqueness of the base in Section 5. For $k = 4$, the same subsection uses a quadrilateral surface in $X_{4,4/3}$, not in the base family; its single smooth-point contraction to Z_4 and Lemma B.5 give $\text{Br}(Z_4) = 0$. Finally, the toric computation in Section B.4 gives $\text{Br}(Z_6) \cong \mathbb{Z}/3$. Altogether,

$$\text{Br}(Z_k) \cong \begin{cases} 0, & 1 \leq k \leq 5, \\ \mathbb{Z}/3, & k = 6. \end{cases}$$

For every surface X in an index-one family $X_{k,d}$, (7) now gives $\text{Br}(X) \cong \text{Br}(Z_k)$. This uses the smooth-point blow-up presentation of each surface, not an assumption that its Brauer group is determined by a general member.

The auxiliary families $B_{1,16/3}$ and $B_{2,8/3}$ are covered directly by Corollary B.4, since their members have one and two singular points, respectively. They also fit the cascade argument: the alternative routes in [7, Remark 9], recalled in Section 5, give a smooth-point blow-up $X' \rightarrow B$ and four floating contractions from X' to Z_k , for $k = 1, 2$ respectively. Reversing those contractions and applying Lemma B.5 gives

$$\text{Br}(B) \cong \text{Br}(X') \cong \text{Br}(Z_k) = 0.$$

Together with the already treated base Z_1 , these account for all higher-index families.

We have now covered every family in the six cascades. The final assertion follows from [19, Corollary 4.8]: the nonzero Brauer group in the six-point cascade obstructs an untwisted adherent decomposition, whereas this Brauer obstruction is absent in the other five cascades. This proves Theorem 1.2. $\square$

C Explicit exceptional collections on the rigid cascade bases

This appendix proves [Proposition 5.4](#) by specifying the initial collections for all six cascades. The models of the singular surfaces come from Corti–Heuberger and [Appendix B](#); the blow-down sequences and the elementary transformation below make their smooth resolutions explicit. These constructions serve a different purpose from the exceptional-lattice computations in [Appendix B](#).

After fixing uniform notation, we treat the six bases in order. [Sections C.1](#) and [C.2](#) begin with $\mathbb{F}_3$ and distinguish the incidence conditions that create a second quotient point. The three- and six-point cases are checked by toric contractions; the four-point case requires an elementary transformation, and the five-point case uses the pentagon construction. Each case ends with its ordered resolution collection, and [Corollary C.3](#) propagates them through a chosen marked index-one cascade.

The following guide records the centers and geometric steps for the models in [Table 1](#). All modifications are ordinary blow-ups at smooth points; “infinitely near” describes the location of a center relative to the starting surface Y_0 listed there.

k	Centers	Key geometric step
1	None	Use the negative section.
2	One point and two infinitely near centers	Create a second (-3) -curve.
3	One torus-fixed point and three infinitely near toric centers	Reverse four boundary contractions.
4	Five distinct points and two infinitely near centers	Perform an elementary transformation.
5	Four plane points, then five vertices (four infinitely near)	Separate the pentagon curves.
6	Four torus-fixed points and four infinitely near toric centers	Contract to a hexagon, then a square.

The individual cases below specify the centers and their order.

We use the following notation throughout. For a sequence of ordinary point blow-ups

$$Y_m \xrightarrow{\beta_m} Y_{m-1} \longrightarrow \cdots \xrightarrow{\beta_1} Y_0, \quad \sigma = \beta_1 \circ \cdots \circ \beta_m,$$

let $D_i \subset Y_i$ be the exceptional curve created by β_i , and set

$$G_i = L\beta_m^* \cdots L\beta_{i+1}^* \mathcal{O}_{D_i}(-1). \quad (9)$$

Thus $G_m = \mathcal{O}_{D_m}(-1)$. If $\langle L_1, \dots, L_t \rangle$ is a full exceptional collection of line bundles on Y_0 , [Theorem 2.5](#) gives the ordered full exceptional collection

$$\langle G_m, \dots, G_1, \sigma^* L_1, \dots, \sigma^* L_t \rangle. \quad (10)$$

In particular, infinitely near centers are allowed; the objects in [\(9\)](#) are not replaced by sheaves on the final strict transforms. The starting collections, recalled in [Section 2](#), are

$$\begin{aligned} \mathcal{B}_{\mathbb{P}^2} &= \langle \mathcal{O}, \mathcal{O}(H), \mathcal{O}(2H) \rangle, \\ \mathcal{B}_{\mathbb{F}_n} &= \langle \mathcal{O}, \mathcal{O}(f), \mathcal{O}(C_0 + nf), \mathcal{O}(C_0 + (n+1)f) \rangle. \end{aligned}$$

Here H is a line class, and C_0, f are the section and fiber classes with $C_0^2 = -n$. We denote the collection on $\tilde{Z}_k$ by $\mathcal{E}_k$, the canonical stack of Z_k by $\mathcal{Z}_k$, and its Ishii–Ueda embedding by Φ_k . In each case, [Corollary 3.6](#) turns $\mathcal{E}_k$ into a stack collection by applying Φ_k and prepending the k local objects. The lengths are recorded in [Table 1](#), so we list only the resolution collections below.

C.1 The one-point base

The one-point base supplies the starting ruled-surface collection without any additional blow-ups, and serves as the reference case for the next construction. The minimal resolution of $Z_1 = \mathbb{P}(1, 1, 3)$ is $\mathbb{F}_3$, with exceptional curve C_0 . No point blow-ups are needed, and we take

$$\mathcal{E}_1 = \langle \mathcal{O}, \mathcal{O}(f), \mathcal{O}(C_0 + 3f), \mathcal{O}(C_0 + 4f) \rangle.$$

C.2 The two-point base

We construct a second quotient point while preserving the original negative section of $\mathbb{F}_3$, thereby specifying the incidence data needed for the two-point base collection. The two-point case of the directed MMP in [7, Section 6] has a branch

$$Z_2 = X_{2,17/3} \longrightarrow T \longrightarrow \mathbb{P}(1, 1, 3), \quad \text{Sing}(T) = \left\{ \frac{1}{3}(1, 1), A_1 \right\},$$

with contractions of types (E.4) and (E.2), respectively, as defined in [7, Theorem 31]. We give a marked reconstruction of this branch and verify its identification with Z_2 independently of a numerical blow-up count.

Proposition C.1. *Choose $q_1 \in \mathbb{F}_3 \setminus C_0$, and let f_0 be its ruling fiber. Blow up q_1 , with exceptional curve D_1 , and then choose $q_2 \in D_1$ away from the strict transform of f_0 . Blow up q_2 , with exceptional curve D_2 . Finally, choose q_3 on the strict transform of D_1 , away from both D_2 and the strict transform of f_0 , and blow it up. Then*

$$\tilde{Z}_2 \simeq Y_3 = \text{Bl}_{q_3} \text{Bl}_{q_2} \text{Bl}_{q_1} \mathbb{F}_3.$$

The two curves contracted to the quotient singularities are the untouched C_0 and the final strict transform A of D_1 .

Proof. Write Y_i for the surface after the first i blow-ups and T_i for the total transform of the class of D_i on Y_3 . Then $T_i T_j = -\delta_{ij}$, and pullbacks from $\mathbb{F}_3$ are orthogonal to all T_i . On Y_2 the strict transform A_2 of D_1 has square -2 . The last center lies on A_2 , but on neither of its two neighboring curves. Consequently

$$A = T_1 - T_2 - T_3, \quad A^2 = -3, \quad C_0 A = 0.$$

The strict transform $F' = f - T_1$ of f_0 has square -1 ; it meets C_0 and A once each. The curves D_2, D_3 remain (-1) -curves, each meeting A once and disjoint from C_0 and F' . The dual graph records these transverse intersections (each edge has intersection multiplicity one):

$$\begin{array}{ccccc} & & & D_2(-1) & \\ & & & | & \\ C_0(-3) & \text{---} & F'(-1) & \text{---} & A(-3) \\ & & & | & \\ & & & D_3(-1) & \end{array}$$

To check that contracting C_0 and A gives the required del Pezzo surface, consider the rational divisor

$$P = -K_{Y_3} - \frac{1}{3}(C_0 + A) = \frac{5}{3}C_0 + 5f - \frac{4}{3}T_1 - \frac{2}{3}T_2 - \frac{2}{3}T_3.$$

The intersection form in the orthogonal total-transform basis gives

$$P^2 = \frac{17}{3}, \quad PC_0 = PA = 0, \quad PF' = \frac{1}{3}, \quad PD_2 = PD_3 = \frac{2}{3}.$$

Every other irreducible curve is the strict transform of a fiber not through q_1 , or of a curve $B \sim aC_0 + bf$ with $a \geq 1$: an irreducible curve on $\mathbb{F}_3$ with $a = 0$ is a fiber, and curves exceptional over $\mathbb{F}_3$ were already listed. Since $B \neq C_0$, nonnegative intersection of distinct irreducible curves gives $BC_0 = b - 3a \geq 0$, hence $b \geq 3a$. Let m_1, m_2, m_3 be its multiplicities at the successive centers. Intersection with f_0 gives $m_1 \leq a$, while nonnegative intersection of its final strict transform with $A = T_1 - T_2 - T_3$ gives $\tilde{B}A = m_1 - m_2 - m_3 \geq 0$, hence $m_2 + m_3 \leq m_1$. Consequently,

$$P \cdot \tilde{B} = \frac{5}{3}b - \frac{4}{3}m_1 - \frac{2}{3}(m_2 + m_3) \geq \frac{5}{3}b - 2a \geq 3a > 0.$$

On a fiber not through q_1 , the intersection is $5/3$. Thus P is nef and big, and its only null curves are C_0, A . The pair $(Y_3, (C_0 + A)/3)$ is klt, so the basepoint-free theorem [20, Section 3.2] makes a sufficiently divisible multiple of P semiample. Its associated contraction has precisely these two disjoint (-3) -curves as exceptional locus and gives a normal surface Z with two $\frac{1}{3}(1, 1)$ -points. The discrepancy formula identifies P with the pullback of the ample divisor $-K_Z$.

There is also no ambiguity about the Fano index. The divisor-class group of Z is the quotient of $\text{Pic}(Y_3)$ by C_0, A , so it is free with basis f, T_2, T_3 . In this quotient

$$-K_Z = 5f - 2T_2 - 2T_3$$

is primitive. The surface therefore has Fano index one, and [7, Theorem 4 and Remark 7] identifies it with Z_2 . Since its exceptional curves have square -3 , Y_3 is its minimal resolution. $\square$

This construction also explains the intermediate A_1 in the MMP. On Y_2 , contracting C_0 and A_2 gives T ; the same nef-divisor argument, using $-K_{Y_2} - C_0/3$, supplies this contraction. Contracting D_2 and then the resulting (-1) -curve D_1 returns $\mathbb{F}_3$. The image of D_3 on Z_2 passes through the quotient point defined by A ; contracting it replaces $A(-3)$ by $A_2(-2)$, which is type $(E.4)$. On T , the image of D_2 passes through the A_1 -point. Its resolved configuration is $D_2(-1) - A_2(-2)$; the two successive smooth blow-downs contract this configuration to the smooth point q_1 , giving type $(E.2)$. The composite $Y_2 \rightarrow \mathbb{F}_3 \rightarrow \mathbb{P}(1, 1, 3)$ is constant on both C_0 and A_2 , so it factors through T . Thus the resolutions and the singular surfaces fit into the commutative diagram

$$\begin{array}{ccccc} Y_3 & \xrightarrow{\beta_3} & Y_2 & \xrightarrow{\beta_1 \circ \beta_2} & \mathbb{F}_3 \\ \downarrow & & \downarrow & & \downarrow \\ Z_2 & \longrightarrow & T & \longrightarrow & \mathbb{P}(1, 1, 3). \end{array}$$

Reading the upper row in reverse, the configurations over singular points evolve as

$$C_0(-3) \rightsquigarrow C_0(-3) \sqcup A_2(-2) \rightsquigarrow C_0(-3) \sqcup A(-3).$$

The original quotient point is untouched; the inverse MMP creates the second one. With $\sigma_2 : Y_3 \rightarrow \mathbb{F}_3$ the composite, the seven-object collection is

$$\mathcal{E}_2 = \langle G_3, G_2, G_1, \sigma_2^* \mathcal{O}, \sigma_2^* \mathcal{O}(f), \sigma_2^* \mathcal{O}(C_0 + 3f), \sigma_2^* \mathcal{O}(C_0 + 4f) \rangle.$$

A threefold blow-up of $\mathbb{F}_3$ can also occur in the one-point cascade; here the two later centers on the first exceptional curve create the second (-3) -curve A .

C.3 The three-point base

For the three-point base, we turn the toric resolution of [Section B.3](#) into a marked four-step blow-up of $\mathbb{F}_3$, so that its initial collection is specified by boundary intersections. The fan of Z_3 has rays

$$v_1 = (1, 0), \quad v_2 = (-1, 3), \quad v_3 = (-2, 3), \quad v_4 = (-1, 0), \quad v_5 = (0, -1),$$

and its minimal resolution inserts

$$u_{12} = (0, 1), \quad u_{23} = (-1, 2), \quad u_{34} = (-1, 1).$$

In the boundary order $V_1, E_1, V_2, E_2, V_3, E_3, V_4, V_5$, the self-intersections are

$$(0, -3, -1, -3, -1, -3, -1, 0).$$

We contract, in this order, the successive images of V_4, V_3, E_3, E_2 . The following table lists the neighboring rays at the actual stage of each contraction:

curve	neighboring rays	ray relation
V_4	u_{34}, v_5	$v_4 = u_{34} + v_5$
V_3	u_{23}, u_{34}	$v_3 = u_{23} + u_{34}$
E_3	u_{23}, v_5	$u_{34} = u_{23} + v_5$
E_2	v_2, v_5	$u_{23} = v_2 + v_5$

In every row the determinant of the two neighbors is one. The ray-sum criterion therefore verifies that the curve being removed is a (-1) -curve and that its contraction is a smooth toric blow-down [9, Theorem 10.4.4]. The remaining fan has rays v_1, u_{12}, v_2, v_5 , with boundary squares $(0, -3, 0, 3)$; it is the fan of $\mathbb{F}_3$, with negative section corresponding to u_{12} .

Reversing these four contractions prescribes the centers: insert u_{23} in the cone $\langle v_2, v_5 \rangle$, then u_{34} in $\langle u_{23}, v_5 \rangle$, then v_3 in $\langle u_{23}, u_{34} \rangle$, and finally v_4 in $\langle u_{34}, v_5 \rangle$. Each center is the intersection of the two indicated boundary curves. This gives $\sigma_3 : \tilde{Z}_3 \rightarrow \mathbb{F}_3$ and the full exceptional collection

$$\mathcal{E}_3 = \langle G_4, G_3, G_2, G_1, \sigma_3^* \mathcal{O}, \sigma_3^* \mathcal{O}(f), \sigma_3^* \mathcal{O}(C_0 + 3f), \sigma_3^* \mathcal{O}(C_0 + 4f) \rangle.$$

The creation stages for G_1, G_2, G_3, G_4 correspond respectively to the rays u_{23}, u_{34}, v_3, v_4 ; in particular the first two objects are understood by derived pullback, not as sheaves on the final (-3) -curves.

C.4 The four-point base

A geometric reconstruction is needed here because the quadrilateral model in [Section B.3](#) lies in $X_{4,4/3}$, not in the rigid base family $Z_4 = X_{4,7/3}$. The following proposition supplies the additional contraction and the change of ruled surface required for its marked blow-up presentation.

Proposition C.2. *Let $S = \text{Bl}_{p_1, \dots, p_5} \mathbb{P}^2$, with the points general, and let $H, Q_1, \dots, Q_5$ be its standard divisor classes. Set*

$$\begin{aligned} C_1 &= Q_1, & C_2 &= H - Q_1 - Q_2, \\ C_3 &= Q_2, & C_4 &= 2H - Q_1 - Q_2 - Q_3 - Q_4 - Q_5. \end{aligned}$$

Blow up the vertices $r_i = C_i \cap C_{i+1}$, cyclically, with exceptional curves R_i , obtaining Y . The strict transform of the line tangent at p_1 to the conic through the five points is a (-1) -curve

$$D = H - Q_1 - R_4.$$

Its contraction $Y \rightarrow Y'$ identifies Y' with $\tilde{Z}_4$. Moreover, an elementary transformation at r_4 gives an explicit seven-point blow-up presentation $\sigma_4 : \tilde{Z}_4 \rightarrow \mathbb{F}_2$. Its centers, in order, are the transported points $p_2, p_3, p_4, p_5, r_1, r_2, r_3$: the first five are distinct on $\mathbb{F}_2$, and the last two are infinitely near p_2 .

Proof. All divisor classes on Y in the calculations below are total pullbacks, except where strict transforms are explicitly specified. The four (-1) -curves C_i form a cycle with distinct vertices, so their strict transforms

$$E_i = C_i - R_{i-1} - R_i$$

satisfy $E_i^2 = -3$ and $E_i E_j = 0$ for $i \neq j$. The contraction $Y \rightarrow X_{4,4/3}$ is the 4-gon construction of [7, Section 3.5].

The conic through five general points is smooth. Its tangent line at p_1 contains none of $p_2, \dots, p_5$, by Bézout, and meets Q_1 at r_4 after the first blow-up. This point is different from r_1 , the direction of the secant through p_2 . Thus D is indeed the strict transform of an irreducible smooth rational curve, rather than just a divisor class. In the orthogonal blow-up basis we have

$$D^2 = -1, \quad (-K_Y)D = 1, \quad DE_i = 0 \quad (1 \leq i \leq 4).$$

These distinct irreducible curves are therefore disjoint. The image of D on $X_{4,4/3}$ lies in the smooth locus and is a floating (-1) -curve. Its contraction is a log del Pezzo surface with four unchanged $\frac{1}{3}(1,1)$ -points and degree $7/3$, hence is Z_4 by the classification in Section 5. Contracting D on Y leaves the four E_i unchanged and gives its minimal resolution Y' .

To identify the starting ruled surface, first blow up only p_1 . This gives $\mathbb{F}_1$, with negative section Q_1 and ruling class $H - Q_1$. The point r_4 is already specified on Q_1 by the tangent direction of the conic; its definition does not require blowing up the other four points first. Put $W = \text{Bl}_{r_4} \mathbb{F}_1$. The strict transform of that ruling fiber is D . Contracting it is the elementary transformation of the ruled surface at a point on its negative section. The ruling descends to a $\mathbb{P}^1$ -bundle over $\mathbb{P}^1$, and the image of the strict transform of Q_1 is a section of square -2 , since it is disjoint from D . The classification of ruled surfaces therefore identifies the new surface with $\mathbb{F}_2$ [14, Chapter V, Section 2].

It remains to account for the marked centers, rather than just their number. On W , perform the blow-ups at p_2, p_3, p_4, p_5 , then at r_1, r_2, r_3 . All these centers, including the two directions r_2, r_3 on the exceptional curve over p_2 , lie away from the strict transform of D : the tangent line avoids the four points, and $r_1 \neq r_4$. These seven blow-ups commute with the contraction of D . More explicitly, the isomorphism

$$W \setminus D \simeq \mathbb{F}_2 \setminus \{t\},$$

where t is the image of D , transports these centers and their successive infinitely near directions to $\mathbb{F}_2$. Blow up the images of p_2, p_3, p_4, p_5 , followed by the transported r_1, r_2, r_3 . The transported r_1 lies on the negative section of $\mathbb{F}_2$; unlike the one-point cascade construction, this base model does not require every center to avoid that section. This is the asserted morphism σ_4 , and proves the isomorphism by commuting the actual modifications. $\square$

Using precisely this order of seven centers in (9), we obtain

$$\mathcal{E}_4 = \langle G_7, \dots, G_1, \sigma_4^* \mathcal{O}, \sigma_4^* \mathcal{O}(f), \sigma_4^* \mathcal{O}(C_0 + 2f), \sigma_4^* \mathcal{O}(C_0 + 3f) \rangle.$$

Here C_0 is the negative section on the new $\mathbb{F}_2$.

C.5 The five-point base

The five-point base needs no further contraction beyond the pentagon construction: its nine blow-ups directly provide the marked plane model and hence the initial Orlov collection. On the degree-five del Pezzo surface $S_5 = \text{Bl}_{p_1, \dots, p_4} \mathbb{P}^2$, take the pentagon from Section B.3:

$$\begin{aligned} C_1 &= Q_1, & C_2 &= H - Q_1 - Q_2, & C_3 &= Q_2, \\ C_4 &= H - Q_2 - Q_3, & C_5 &= H - Q_1 - Q_4. \end{aligned}$$

Each curve has square -1 , adjacent curves intersect once, and non-adjacent curves are disjoint. Blow up the five distinct vertices $r_i = C_i \cap C_{i+1}$, with exceptional curves R_i . Since exactly two vertices lie on each C_i , the strict transforms

$$E_i = C_i - R_{i-1} - R_i$$

have square $-1 - 2 = -3$. Blowing up the unique intersection of any adjacent pair separates it, so $E_i E_j = 0$ for $i \neq j$. By the 5-gon construction of [7, Section 3.5], their contraction belongs to $X_{5,5/3}$. Uniqueness of this base identifies the contraction with Z_5 , and consequently

$$\tilde{Z}_5 \simeq \text{Bl}_{r_1, \dots, r_5} \text{Bl}_{p_1, \dots, p_4} \mathbb{P}^2.$$

Here the vertex blow-ups are on S_5 , not at nine distinct points of the plane: r_1, r_5 lie over p_1 , r_2, r_3 lie over p_2 , and r_4 maps to a fifth distinct plane point. Order the centers as $p_1, \dots, p_4, r_1, \dots, r_5$, and write σ_5 for their composite. The twelve-object Orlov collection is

$$\mathcal{E}_5 = \langle G_9, \dots, G_1, \sigma_5^* \mathcal{O}, \sigma_5^* \mathcal{O}(H), \sigma_5^* \mathcal{O}(2H) \rangle.$$

C.6 The six-point base

For the six-point base, the alternating toric boundary gives a route through the degree-six del Pezzo surface to $\mathbb{F}_0$, making the twelve-object resolution collection explicit despite the coarse-surface Brauer obstruction. For the toric model of Z_6 in Section B.4, the rays are

$$\begin{aligned} v_1 &= (1, 0), & v_2 &= (-1, 3), & v_3 &= (-2, 3), \\ v_4 &= (-1, 0), & v_5 &= (1, -3), & v_6 &= (2, -3), \end{aligned}$$

and the resolution inserts

$$\begin{aligned} u_1 &= (0, 1), & u_2 &= (-1, 2), & u_3 &= (-1, 1), \\ u_4 &= (0, -1), & u_5 &= (1, -2), & u_6 &= (1, -1). \end{aligned}$$

Its boundary is $V_1, E_1, \dots, V_6, E_6$, with $V_i^2 = -1$ and $E_i^2 = -3$. We first contract $V_1, \dots, V_6$, in that order. At every stage, the neighbors of the ray v_i being removed are still u_{i-1}, u_i , and

$$u_{i-1} + u_i = v_i, \quad \det(u_{i-1}, u_i) = 1.$$

Thus every contraction is a smooth toric blow-down. Equivalently, the V_i are pairwise disjoint, so their self-intersections remain -1 until they are contracted. After any initial set I of these contractions, the self-intersection of the image of E_j is

$$-3 + \#(I \cap \{j, j+1\}),$$

with cyclic indices. After all six, the fan consists of the u_i and every boundary curve has square -1 . This is the smooth toric del Pezzo surface of degree six: its anticanonical divisor intersects each invariant curve positively, and its square is six.

Next contract the opposite boundary curves with rays u_1 and u_4 . They are disjoint, and their ray relations are

$$u_1 = u_6 + u_2, \quad u_4 = u_3 + u_5, \quad \det(u_6, u_2) = \det(u_3, u_5) = 1.$$

Both are therefore (-1) -curves at the moment of contraction. The remaining rays are $u_2, u_3, -u_2, -u_3$, and $\det(u_2, u_3) = 1$. In this lattice basis the fan is the square fan of $\mathbb{F}_0 = \mathbb{P}^1 \times \mathbb{P}^1$; this checks the endpoint directly, not just its degree.

Reversing the sequence prescribes eight toric point blow-ups: first insert u_4 between u_3, u_5 , then u_1 between u_6, u_2 , and then insert $v_6, v_5, \dots, v_1$, each between u_{i-1}, u_i . The insertions of u_4, u_1, v_6, v_3 blow up the four original fixed points of $\mathbb{F}_0$; the remaining four centers are infinitely near those producing u_4 and u_1 . Let $\sigma_6 : \tilde{Z}_6 \rightarrow \mathbb{F}_0$ be their composite. With f_1, f_2 the two ruling classes on $\mathbb{F}_0$, the resulting twelve-object collection is

$$\mathcal{E}_6 = \langle G_8, \dots, G_1, \sigma_6^* \mathcal{O}, \sigma_6^* \mathcal{O}(f_1), \sigma_6^* \mathcal{O}(f_2), \sigma_6^* \mathcal{O}(f_1 + f_2) \rangle.$$

The resulting stack collection has length eighteen; the nonzero Brauer group of Appendix B obstructs untwisted adherent KKS descent to the coarse surface.

The six constructions establish Proposition 5.4. Together with the cascade procedure already described in Section 5, the following corollary proves Theorem 1.1. We use $A_j^{(k)}$ for arbitrary entries of a base collection, reserving G_i for blow-up objects.

Corollary C.3. *Let X belong to an index-one family $X_{k,d}$, and put $s = K_{Z_k}^2 - K_X^2$. Choose a marked Corti–Heuberger presentation by s distinct smooth points of Z_k , and let $\gamma : \tilde{X} \rightarrow \tilde{Z}_k$ be the induced map of minimal resolutions. Write the base collection constructed above as*

$$\mathcal{E}_k = \langle A_1^{(k)}, \dots, A_{N_k}^{(k)} \rangle,$$

with N_k as in Table 1, and let $D_1, \dots, D_s$ be the disjoint exceptional curves of γ . Set $H_i = \mathcal{O}_{D_i}(-1)$; distinctness follows from (3). The derived-pullback convention (9) is needed only inside the base collections, whose centers may be infinitely near. Then

$$D^b(\mathrm{coh} \tilde{X}) = \langle H_s, \dots, H_1, L\gamma^* A_1^{(k)}, \dots, L\gamma^* A_{N_k}^{(k)} \rangle$$

is a full exceptional collection. If $\mathcal{X}$ is the canonical stack and Φ its Ishii–Ueda embedding, the ordered collection

$$D^b(\mathrm{coh} \mathcal{X}) = \langle \mathcal{E}_{p_1}, \dots, \mathcal{E}_{p_k}, \Phi(H_s), \dots, \Phi(H_1), \Phi(L\gamma^* \mathcal{E}_k) \rangle$$

is full exceptional, of length $N_k + s + k = 12 - K_X^2 + 4k/3$, the specialization of Corollary 3.6 with $n_i = 3$ and $r = k$.

Remark C.4 (Marked presentations and dominance). The memberwise presentations of [7, Corollary 8(1)–(6)] have distinct smooth centers by (3); they suffice for all cascade collections in Theorem 1.1. In particular, every hypersurface satisfying Theorem 1.3 has the stated collection, Euler form, and coarse decomposition. Appendix A adds a comparison of generality: general hypersurfaces admit general centers, and general centers give general hypersurfaces. It controls automorphisms and finite ambiguity when comparing these two parameter spaces.

For other families, an analogous comparison with a specified equation parameter space or an alternative marked model requires its own classifying map and fiber analysis. It is this comparison, not the existence of distinct cascade centers or their categorical consequences, that remains beyond the present work. The infinitely near centers in Appendix C belong to the fixed base resolutions, not to the additional cascade blow-ups. These generality questions are independent of constructing simultaneous adherent blocks; Brauer vanishing alone still does not construct such blocks for $k = 2, 4, 5$.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author used ChatGPT (OpenAI) to assist with language editing, organization, and manuscript formatting. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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